



Iconicity and expressiveness of grammatical tones targeting ideophones in Mwaghavul

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Abstract

All tonally derived verbs in Mwaghavul have ideophonic bases and are derived through one of two lexically determined alternations. In the first alternation, every tone-bearing unit (TBU) of the ideophone surfaces with a mid tone. In the second alternation, every nonfinal TBU of ideophones surfaces with a mid tone, while the final TBU surfaces with a high tone. To express pluractionality, both types of ideophones are reduplicated and follow the second pattern: every TBU of the reduplicant bears a mid tone, and every TBU of the base bears a high tone. While verbs derived through mid-tone overwrite share a tonal melody with underived verbs, only derived verbs have a mid–high melody. Beyond verb formation, the tonal alternations introduce various meanings, conveying the distinguishability of identical elements. We propose that the alternations are triggered by lexically specific allomorphs of a verbaliser, whose exponents are mid and mid–high tonal melodies, respectively. The restriction of the mid–high melody to derived verbs reflects systematicity. For the derived pluractional verbs, the tonal dissimilarity between a reduplicant and its base depicts the distinguishability of otherwise identical elements. We consider the implications of the patterns for linguistic theory, particularly the relationship between grammatical integration and expressiveness.

Keywords Grammatical tone · Iconicity · Systematicity · Chadic · Ideophones · Grammatical integration · Disharmony

1 Introduction

We present evidence that grammatical tone can target ideophones and that iconicity can motivate the realisation of such a tone. Ideophones, defined as “an open lexical

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class of marked words that depict sensory imagery”, are a universal feature of human language (Dingemanse 2019:16). In the linguistic literature, a recurrent theme about ideophones is their marked grammatical status. For example, most documented cases of ideophones have found that they are rarely subject to the same phonotactic restrictions as other word classes and typically have phonetic properties that are absent elsewhere in the grammar. They also have been shown to occur mostly outside predicate structures and to lack inflectional and derivational affixes. As a result, ideophones are considered to be less integrated into core grammar (Voeltz and Kilian-Hatz 2001; Körtvélyessy and Štekauer 2024). Their meanings tend to be affective, ineffable and dependent on the utterance situation (Voeltz and Kilian-Hatz 2001). The fact that the meanings of ideophones are often tied to the time and place of utterance is referred to as *nondisplaceability* (Potts 2007:169). Due to their depiction of sensory imagery, the form-meaning mapping of ideophones tends to involve iconicity, which is a resemblance between linguistic form and referential meaning (Perniss et al. 2010; Dingemanse et al. 2015). As a notable example of iconicity, onomatopoeic ideophones for a rooster’s crow across languages are perceptually similar to an actual rooster’s crow (e.g., *cock-a-doodle-doo* in English, *kokekoko* in Japanese, *kukuRiku* in Hebrew and *kukuluukuu* in Kisi (Childs 1994:189)). Partly due to the longstanding view that iconicity and expressive morphology occur at the margins of grammar (de Saussure 1916; Hockett 1960; Zwicky and Pullum 1987), ideophones have received little theoretical attention until recently.

Contrary to the traditional view that iconicity plays no role in grammar, evidence from more recent descriptive and experimental studies of ideophones and other expressive words indicates that iconicity bootstraps language acquisition in both children and adults (Imai and Kita 2014; Lockwood and Dingemanse 2015), facilitates the phonological categorisation of semantically related words (Akita et al. 2024), and contributes to grammaticalisation processes involved in the emergence of new lexical items (Childs 1996; Andrason and Bernd 2023; Heine 2023). While ideophones tend to occur outside core grammar, studies suggest that they can nevertheless still be grammatically integrated. The degree to which ideophones are grammatically integrated into predicate structure inversely correlates with their level of expressiveness (Dingemanse and Akita 2017; Andrason and Bernd 2023). This means that grammatically integrated ideophones may lose their marked properties, resistance to affixation and semantic ineffability. These recent findings, among others, support the more contemporary perspective that arbitrariness and iconicity play complementary roles in grammar (Perniss et al. 2010; Dingemanse et al. 2015).

Despite progress in the field, iconicity is rarely considered in synchronic analyses of sound patterns, especially those involving tone. Of particular relevance to the present work is the absence of data on ideophones in the study of grammatical tone, defined as a tonal operation that is “restricted to a specific morpheme or construction... and not attributable to the general tonal phonology” (Lionnet et al. 2022:386). An example of grammatical tone from Berom (Plateau, Nigeria) involves the pluralisation of nouns by overwriting every lexical tone of a noun with a high tone, such that the nouns [p]è] ‘thing’, [fùgù] ‘speck’, and [tʃògòt] ‘tree’ have [p]é] ‘things’, [fùgù] ‘specks’, and [tʃògòt] ‘trees’ as their respective plural forms (Ukaegbu et al. 2017:238). The underrepresentation of ideophones from tone studies is not limited to

grammatically defined tone replacement but also extends to tone polarity, a dissimilatory process that prohibits a sequence of identical tones on adjacent tone-bearing units (Leben 1973; Goldsmith 1976). However, recent cross-linguistic studies have found that tone polarity and other forms of disharmony are also attested in ideophones, even in languages with relatively minimal ideophones (Dingemanse and Thompson 2020; Wivell et al. 2024).

In this work, we present evidence that grammatical tone is another formal grammatical feature that can target ideophones. Considering that grammatical tone is an integral aspect of grammar, another major goal of this study is to explore whether the forms derived from ideophones through grammatical tones still continue to exhibit expressive ideophonic features, and hence can be said to remain within the class of ideophones. The empirical basis of our descriptive and theoretical exploration in this work is the derivation of verbs in Mwaghavul, an understudied Chadic language. All tonally derived verbs in the language have ideophonic bases, with each verb formed through one of two patterns of tonal alternation illustrated by [wùlǎŋ] ‘wide open’ and [kǐdǐŋ] ‘looking gorgeous’ in (1).

(1) Verb formation from ideophones in Mwaghavul

	‘wide open’	‘looking gorgeous’	‘heavy fall’	‘light fall’
a.	wùlǎŋ	kǐdǐŋ	dʒùm	dʒúm
b. Verb	wùlǎŋ	kǐdǐŋ	—	—
c. Verb.PL	wùlǎŋ-wùlǎŋ	kǐdǐŋ-kǐdǐŋ	dʒùm-dʒúm	dʒùm-dʒúm

To derive verbs, the first set of ideophones surfaces with an mid (M) tone on each tone-bearing unit (TBU). To derive verbs from the second set of ideophones, every nonfinal TBU surfaces with an M tone, while the final TBU surfaces with a high (H) tone. Both patterns of tonal alternation are illustrated in (1b). Whether an ideophone surfaces with an M or M-H melody in verb formation is lexically specific, as verbs with similar syllable and segment types may undergo distinct tonal alternations. To derive pluractional verbs, the two classes of ideophones undergo reduplication and an M-H tonal alternation, such that every TBU of the reduplicant (i.e., the leftmost copy) surfaces with an M tone, and every TBU of the base (i.e., the rightmost copy) surfaces with an H tone, as shown in (1c).¹ In the case of pluractional verbs, there is no lexical distinction in tonal pattern as seen in the verbal bases of (1b). The verbs derived through the M-H tonal melody are also phonologically exceptional, as they are the only verbs with this tone pattern in the language. In addition to verb formation, the tonal alternations also introduce various meanings, which generally convey the distinguishability of identical elements. For example, the verbs derived from the ideophone [kǐdǐŋ] ‘looking gorgeous’ in (1b–c) express that an entity (e.g., a person, animal, chair, etc.) is distinctively gorgeous when compared to other similar entities (or events). The exact meaning of the derived verbs is difficult to capture without considering their ideophonic bases and the utterance situation. In other words, the

¹ We cannot determine which part is the base and which is the reduplicant. Given that prefixing reduplicants are more common crosslinguistically (Hurch 2005), as well as in Mwaghavul (see Erin and Fwangwar 2024), we refer to the leftmost copy as the reduplicant and the rightmost as the base. This categorisation is purely descriptive.

verb-forming tonal alternations modify the meaning of the ideophonic stems with varying and context-dependent meanings. As with many ideophones in the language, the derived verbs have positive or negative affective meanings. The derived verbs morphosyntactically pattern like underived verbs in the language, as they are integrated into predicate structure and bear inflectional affixes (see §3).

The tonal alternations of verb formation in Mwaghavul are consistent with grammatical tone. In §4, we present an account of the alternations based on the framework of Rolle (2018). Our hypothesis in this work is that the alternations are triggered by verbalisers with only M and M-H tone melodies as their respective exponents. Due to having similar meanings and their unpredictable distribution, the verbalisers are considered lexically specific allomorphs of the same morpheme. The verbaliser with an M tone targets only unreduplicated ideophones, while the verbaliser with an M-H melody targets unreduplicated and reduplicated ideophones. For example, the ideophone [wùlǎ] ‘wide open’ undergoes an M tonal overwrite, and the ideophone [kǐdǐŋ] ‘looking gorgeous’ undergoes an M-H tonal overwrite in the derivation of singular verbs, surfacing as [wūlǎ] ‘to be distinguishably wide open’ and [kǐdǐŋ] ‘to be distinguishably gorgeous’, respectively, as seen in (1b). In the derivation of pluractional verbs, the ideophones are reduplicated and bear an M-H melody, surfacing as [wūlǎ]-[wúlǎ] ‘to be distinguishably wide open (pl)’ and [kǐdǐŋ]-[kǐdǐŋ] ‘to be distinguishably gorgeous (pl)’, as seen in (1c). As each tone of the M-H and M verbalisers never crosses the boundary of a root morpheme in their realisation, the host of each tone of the verbalisers is considered a root morpheme. In the derivation of pluractional verbs, the tones of the M-H verbalisers could have been realised on either the ideophonic base or its reduplicant. Instead, the M tone of the M-H verbaliser is realised on every TBU of the reduplicant, while the H tone is realised on every TBU of the base.

To understand the motivation for the particular association of the M tone of the M-H verbaliser with the reduplicant and the H tone with the base, we compare the phonological properties of the derived verbs with those of other ideophones with similar meanings across languages. Crosslinguistically, ideophones expressing distinguishable identity tend to be reduplicated forms, such that a reduplicant and its base have different values for a phonological feature. For instance, the base and reduplicant of many English ideophones expressing distinguishable identity contain vowels with opposite values for the feature [high], such as [zɪɡ-zæɡ] ‘zigzag’ and [klɪp-klɒp] ‘clip-clop’. Consequently, this pattern of reduplication is also known as ablaut reduplication (Wivell et al. 2024) and nonidentical reduplication (Lee 1992). Studies suggest that expressing the distinguishability of identical elements through corresponding segments of a reduplicant and its base bearing different values for a phonological feature reflects a pattern of form–meaning resemblance (Dingemanse and Thompson 2020; Yliniemi 2024; Akinbo 2025). By extension, the base and reduplicant of the derived pluractional verbs bearing different tones is consistent with this pattern of iconicity. In addition, the base and reduplicant of the ideophones bearing different tones can also be considered a pattern of tone polarity. This indicates that the derived pluractional verbs in the language present evidence for tonal polarity motivated by iconicity.

The pattern of verb derivation in Mwaghavul indicates that grammatical tone targets not only nonideophonic words but also ideophones. The tonal alternation of the

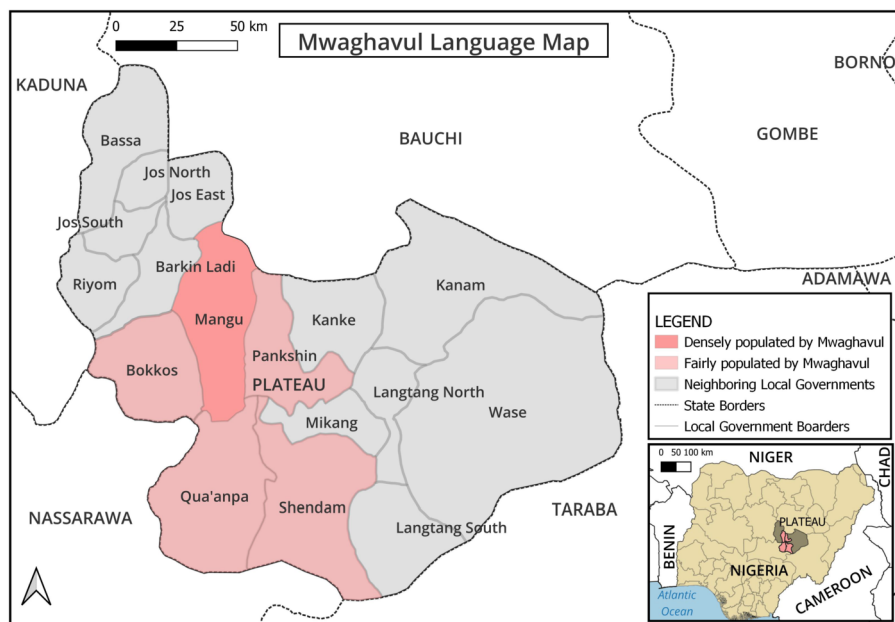


Fig. 1 Mwaghavul speakers and their native settlements in Plateau State, Nigeria

derived pluractional verbs also indicates that iconicity is deeply integrated into the phonological grammar of Mwaghavul, as the tonal distinction between ideophones and their reduplicants diagrammatically depict the distinguishability of identical elements. The affective, ineffable, and context-dependent meanings, as well as the iconicity of the derived pluractional verbs, are consistent with the expressive properties of ideophones across languages. Researchers have argued that the grammatical integration of ideophones inversely correlates with their degree of expressiveness (Dingemanse and Akita 2017; Andrason and Bernd 2023). However, the expressive properties of the derived verbs in Mwaghavul suggest that grammatical integration does not necessarily reduce expressiveness. Other implications of the verb derivation in the language are discussed in §5. In §6, we present our conclusions. As background to the discussion in this work, the following section presents the ethnolinguistic context and aspects of Mwaghavul grammar that are relevant to this study.

2 Language background

Mwaghavul is an A3 West Chadic, Afro-Asiatic language with more than 700,000 speakers (Crozier and Blench 1992; Barnabas 2011). The language is spoken in the Local Government Areas (LGA) of Qua'an Pan, Bokkos, Shendam, Pankshin and predominantly in Mangu of Plateau State, Nigeria. Mwaghavul people are also found in Kaduna and Nasarawa. Their geographical locations are plotted in Fig. 1, which was made in QGIS (Graser et al. 2025). Nyang et al. (2013) note that educated native

Table 1 Phonemic plain consonants in Mwaghavul

	Bilabial		Labio-dental		Alveolar		Post-alveolar		Palatal		Labial-palatal		Velar		Labial-velar		Glottal	
Plosive	p	b			t	d							k	g			ʔ	
Affricate									tʃ	dʒ								
Implosive		ɓ				ɗ												
Fricative			f	v	s	z	ʃ	ʒ					y				h	
Nasal		m				n				ɲ				ŋ				
Trill						r												
Appr. <i>Lateral</i>						l				j					w			

Mwaghavul people hardly speak the language among themselves or to their children, preferring English or Hausa, which is the lingua franca of Northern and Central Nigeria. As a result, Nyang et al. (2013) argue that the language is endangered. Nyang et al., however, believe that recent efforts at documenting and promoting the language have kept the rate of its endangerment in check.

As an additional effort to document and describe aspects of Mwaghavul grammar, the present work focuses on the derivation of verbs from ideophones through two patterns of tonal alternation. Unless stated otherwise, this study is based on data from Fwangwar (2018), Blench (2011, 2013), and the intuitions of the second author, who is a speaker of the language. The data presented are in phonetic transcription and fully marked for tone. Before turning to the main focus of this work, we briefly discuss relevant aspects of Mwaghavul phonology and morphosyntax. Given that the language is underdocumented, the background discussion necessarily includes a certain amount of detail.

2.1 Segments and syllable structure

Mwaghavul contrasts twenty-seven plain consonants, comprising seven plosives, two implosives, four nasals, eight fricatives, two affricates, three approximants, and one trill (Fwangwar 2018). Similar to other Chadic languages (Newman 2007; Hellwig 2011; Akinbo and Bulkaam 2024a), the two implosives in Mwaghavul are bilabial [ɓ] and alveolar [ɗ]. In Table 1, we present the plain contrastive consonants in the language. The language also has labialised and palatalised stops and fricatives, which are not included in the chart. Throughout this work, the labialised and palatalised consonants are represented as digraphs, such as [gw] and [gj].

The language contrasts six short vowels and their respective long counterparts, similar to other A3 West Chadic languages (Hellwig 2011; Akinbo and Bulkaam 2024a). We present the contrastive vowels in Mwaghavul in Table 2. The contrast between short and long vowels is illustrated with the minimal pairs in (2).

Table 2 Phonemic vowels in Mwaghavul

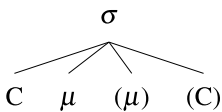
	Front		Central		Back	
High	i	i:	ɨ	ɨ:	u	u:
Mid	ɛ	ɛ:			ɔ	ɔ:
Low			a	a:		

(2) Vowel length contrast

fīt	‘grass’	fīt	‘pounding’
kīn	‘uncle’	kī:n	‘salt’
pūn	‘eject’	pū:n	‘father’
wét	‘spend the day’	wét	‘both’
tòk	‘greet’	tók	‘neck’
pát	‘sheath’	pát	‘olive’

A long or short vowel with an onset consonant can form a syllable in Mwaghavul. The syllable may bear an optional coda consonant, but an onset is obligatory. There are no onset or coda consonant clusters. In addition to vowels, a nasal consonant can form a syllable on its own, but in this case the syllable does not bear an onset or coda.

(3) Syllable structure in Mwaghavul



We propose the syllable structure in (3) for Mwaghavul within the framework of mora theory (Hayes 1989; Hyman 1985). In this theory, a short vowel or syllabic nasal projects one mora, while a long vowel projects two morae. Whether coda consonants in the language are moraic remains a question for further research.

2.2 Tone and tone-bearing unit

Mwaghavul is a tone language like all documented Nigerian languages, except Fulfulde (Williamson 1984). The language contrasts three level tones—high (H), mid (M), and low (L)—as well as an LH rising tone. For an acoustic description of the tones, the second author spoke and hummed monosyllabic Mwaghavul words three times each, comprising nine words with H tone, eight with L tone, ten with M tone and five with rising tone. Following Akinbo and Bulkaam (2024a), we combined the measurements from hummed and spoken words to reduce the effect of the intrinsic F0 of consonants and vowels. The result in Fig. 2 shows that the four tones have distinct F0 heights and trajectories. The acoustic distinction is also consistent with our perceptual impression of the tones.

In Table 3, we illustrate the tone contrasts and tonal combinations using monosyllabic and bisyllabic words. The table shows that all tonal co-occurrences are attested, except for root morphemes with a sequence of M-LH tones. As we will see in §3, the only verbs with a sequence of M-H tones are derived ones, but this tone sequence is attested in both derived and underived words of other word classes.

Fig. 2 Lexical tones in Mwaghavul based on hummed and spoken words

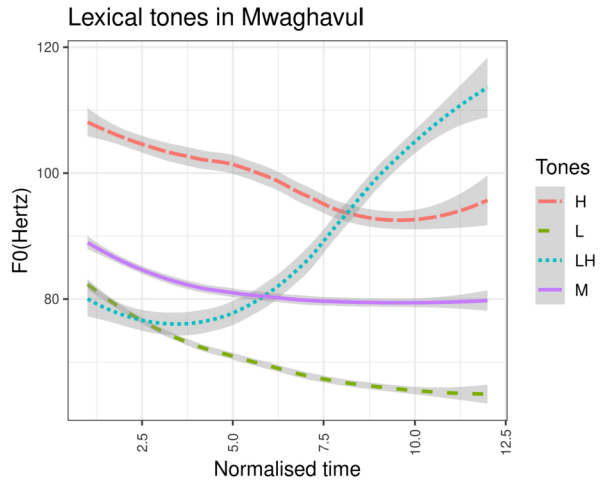


Table 3 Cooccurrence of tones in Mwaghavul root morphemes

			Second tone			
			H	M	L	LH
First tone	H	kún 'three'	jójóm 'firewood'	tímā: 'tobacco'	súlè 'shilling'	kídòm 'hornless ram'
	M	kūn 'rub together'	nānī: 'pygmy mouse'	nīyìn 'mother'	gārvùm 'grasshopper'	
	L	kùn 'big'	dùmú 'hyena'	ràndɔŋ 'cow'	kùlùk 'bag'	tùlù 'home'
	LH	ďũŋ 'river'	dʒǎyám 'be proud'	bīrɪŋ 'horse'	gɔ̌rò 'tree species'	nǎdɔŋ 'new bride'

The discussion in the previous subsection indicates that a syllable in Mwaghavul is either monomoraic or bimoraic. In (4), we examine the distribution of tones across different syllable types.

(4) Syllable types and the distribution of tones

- a. sò 'to move away' tàŋ 'to count'
- sē 'to be sharp' lēk 'to fight'
- sé 'eating' tál 'to strip off a leaf'
- dǒǐ 'in the past' gās 'to share'
- b. là: 'to be fruitful' tà:n 'to beat'
- wā: 'to depart' fē:l 'to peel off'
- dé: 'to be absent' sé:t 'to buy'
- dǎ: 'father' gǒ:r 'bachelor'

As shown in (4), the LH contour and three level tones can be associated with monomoraic as well as bimoraic syllables, indicating that the distribution of tones is not predictable based on a syllable's moraic structure. Since contour tones can associate with both monomoraic and bimoraic syllables, the syllable, rather than the mora,

is considered the tone-bearing unit (TBU) in Mwaghavul. One argument against the mora as the TBU is that no bimoraic syllable bears two distinct contour tones, such as in the hypothetical form C $\check{V}\check{V}$. Having established that the syllable is the TBU in Mwaghavul, we turn to another relevant aspect of the grammar in the next section.

2.3 Ideophones

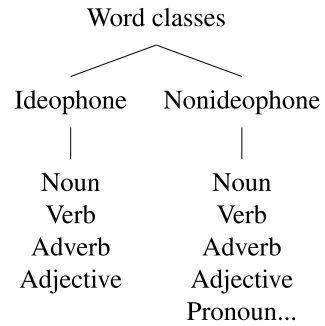
Mwaghavul has a rich inventory of ideophones, “an open class of marked words that depict sensory imagery” (Dingemanse 2019:16). In terms of their segmental properties, the ideophones in the language are not markedly different from nonideophonic words. The only apparent marked property involves the length of words. While most nonideophonic roots are maximally bisyllabic, there is a relatively large number of ideophones with at least three syllables. As open class words, new ideophones are often created by Mwaghavul speakers in expressing both familiar and novel sensory experience. Regardless of whether an ideophone is new or already lexicalised, their phonological forms tend to depict the associated sensory imagery, such that their mode of signification involves mapping the structural components of a real-world entity or event to linguistic form (Emmorey 2014; Thompson et al. 2021). Form-meaning resemblance or iconicity as a defining property of ideophones is based on their depiction of sensory imagery. In (5), we present examples, illustrating a pattern of iconicity.

(5) Magnitude iconicity and tone

- | | | |
|----|-------|--|
| a. | dʒúm | ‘sound of a light object falling in water’ |
| | bíláf | ‘sound of a light person falling down’ |
| | kwá:s | ‘sound of walking fast’ |
| | kjók | ‘sound of dripping lightly’ |
| | kúráf | ‘sound of scratching a body fast’ |
| | wúláf | ‘wide opening of a small mouth’ |
| b. | dʒùm | ‘sound of a heavy object falling in water’ |
| | bíláf | ‘sound of a heavy person falling down’ |
| | kwà:s | ‘sound of walking slowly’ |
| | kjók | ‘sound of dripping heavily’ |
| | kùràf | ‘sound of scratching of a body slowly’ |
| | wúláf | ‘wide opening of a big mouth’ |

Ideophones that express low magnitude, such as the examples in (5a), tend to have an H tone on all their TBUs, while those that express high magnitude, such as the examples in (5b), tend to have an L or M tone on all their TBUs. The examples illustrate a widely attested pattern of iconicity associating high acoustic frequency (e.g., fundamental frequency or second formant) with low magnitude (e.g., smallness, lightness) and conversely low acoustic frequency with high magnitude (e.g., bigness, darkness) (Voeltz and Kilian-Hatz 2001; Hinton et al. 1994; Körtvélyessy and Štekauer 2024). The basis of magnitude iconicity is the inverse correlation between the natural acoustic frequency of a vibrating body and its magnitude (Ohala 1994; Knoeferle et al. 2017; Winter et al. 2021). Just as in the expression of magnitude, other ideophones

Fig. 3 Word classes of ideophonic and nonideophonic words in Mwaghavul



in Mwaghavul are grouped into semantic classes, which may align with an iconic form-meaning mapping (Blench 2013).

Before describing the morphosyntactic distribution of the ideophones, we must first address a fundamental issue concerning word classes. In some languages, ideophones form a lexical class that is distinct from traditional word classes, but in other languages they form a subclass of adverbs (see Akita and Dingemanse 2019:9). In Hausa and Tera, two Chadic languages distantly related to Mwaghavul, there are ideophonic verbs, adverbs and adjectives (Newman 1968). Like these Chadic languages, we propose that traditional open-class words in Mwaghavul can be classified into ideophones and nonideophones, as illustrated in Fig. 3. According to this classification, ideophones occupy the same grammatical classes as nonideophones and so do not constitute a closed lexical class. Supporting this classification is the observation that ideophonic and nonideophonic words in the language exhibit similar morphosyntactic distributions.

The basic word order in Mwaghavul is Subject (S)–Verb (V)–Object (O). As shown in (6), ideophones occur in morphosyntactic environments comparable to those of nonideophonic words.

(6) Syntactic distribution of ideophones

- a. **gìrgìr** pūs ràndōŋ nī
 IDEO hit cow DET
 ‘a heavy-duty vehicle hit the cow’ [as a NOUN]
- b. **gìrgìr** pwās ràndōŋ kí ñmê: nī
 IDEO hit.PL cow CONJ IDEO DET
 ‘a heavy-duty vehicle hit the cow and the goat’ [as a NOUN]
- c. **gìrgìr** pwās ñmú: kí ñmê: nī
 IDEO hit.PL IDEO CONJ IDEO DET
 ‘a heavy-duty vehicle hit the cow and the goat’ [as a NOUN]
- d. má:tò: **gìrgìr** pwās ràndōŋ nī
 motor IDEO hit.PL cow DET
 ‘a heavy-duty vehicle hit the cow multiple times’ [as an ADJECTIVE]
- e. má:tò: nī kí wál kà: sí **gìrgìr**
 motor DET IMPERF cry come:up there IDEO
 ‘there comes the vehicle making **gìrgìr** sounds’ [as an ADVERB]

- f. nā:nlè **múl** mwò:r kó:m
 N. IDEO oil groundnut
 ‘Naanle rubbed/rubs groundnut oil’ [as a VERB]
- g. nā:nlè **m-múl** mwò:r kó:m
 N. FUT-IDEO oil peanut
 ‘Naanle will rub groundnut oil’ [as a VERB]

The subject of the transitive verb [pūs] in (6a) is the onomatopoeic ideophone [gìrgìr], which depicts the sound of a heavy-duty vehicle. Ideophones like [m̩m̩:] ‘goat’ can be conjoined with nonideophonic words or with other ideophones, such as the objects of the verb in (6b–c). Without any phonological alternation or morphological marking, the onomatopoeic ideophone [gìrgìr] seen in (6a–c) functions as an adjective modifying a noun in (6d), and as an adverb modifying the verb in (6e). As illustrated in (6f), some Mwaghavul ideophones are verbs. Like nonideophonic verbs, ideophonic verbs like [múl] can bear inflectional affixes, such as the nasal prefix marking the future tense in (6g).

Regardless of their syntactic position, Mwaghavul ideophones retain their expressive properties. For example, the ideophonic verb [múl] ‘rub’ carries a negative affective meaning, suggesting that the oil was rubbed wastefully or imprecisely. These examples illustrate the more general point that ideophones in Mwaghavul share comparable morphosyntactic distributions with nonideophonic words, without losing their expressive character. In the next section, we show that ideophones form a word class that is distinct from nonideophonic words in Mwaghavul.

3 Expressing distinguishable identity

We describe verb derivation from ideophonic bases in Mwaghavul through two patterns of tonal alternation, beginning with relevant crosslinguistic examples. Across languages, ideophones expressing meanings such as trajectory variation, irregularity, dispersion and nonconformity are typically reduplicated forms in which some segments of the reduplicant and its base have opposite values for a phonological feature. Examples include the vowel disharmony of English ideophones, such as [zɪg-zæŋ] ‘zigzag’ and [dɪli-dæli] ‘dillydally’ (Wallace 2019). Similar patterns occur in Central Tibetan (e.g., [ras-ros] ‘different kinds of furniture’ and [tshab-tshub] ‘helter-skelter’, Yliniemi 2024) and Basque (e.g., [plisti-plasta] ‘splish-splash’ and [zipirti-zaparta] ‘left, right, and centre’, Ibarretxe-Antuñano 2017). This pattern of reduplication is also referred to as nonidentical or ablaut reduplication (Lee 1992; Wivell et al. 2024). In tone languages, the form–meaning mapping extends beyond segmental features to tone. This is evident in ideophones from Bini (e.g., [rúrùrú] ‘staggering’, Wescott 1973), Dagaare (e.g., [gòlòŋ-gólòŋ] ‘zigzag’, Akinbo and Angsongna 2025), Hausa (e.g., [wàndàt-wándát] ‘zigzag’, Sambo 2016:74), Igala (e.g., [ládǝ̄-làdǝ̄] ‘moving aimlessly’, Ejeba 2016:123), Pichi (e.g., [lòk-lók] ‘constantly closing every place’, Yakpo 2019:96), Siwu (e.g., [còkwé-còkwé] ‘irregular sawing motion/sound’, Dingemanse 2011b:135) and Yoruba ([rédèrèdè] ‘useless’, Fordyce 1983:270).

These ideophones are often categorised as denoting “irregularity” (Childs 1994; Dingemanse 2011b; Akinbo 2025), but not all of their meanings are compatible with

this semantic characterisation (Fordyce 1983; Ibarretxe-Antuñano 2017). Based on evidence from five Himalayan languages, Yliniemi (2024) describes the meanings of such ideophones as conveying the idea that parts or instances of an event or entity are similar yet distinguishable. This general meaning aligns with the notion in classical physics that identical particles are distinguishable elements based on their physical, temporal, or spatial properties (Dieks and Lubberdink 2022). Reflecting this parallel, Akinbo and Chiemezie (2024b) label the meaning of these ideophones as ‘distinguishability’ or ‘distinguishable identity’.

Expressing the distinguishability of identical elements through a base-reduplicant composite with dissimilar but aligned feature values constitutes a diagrammatic depiction of real-world elements with linguistic form (Dingemanse 2011a). A phonological form and its reduplicant bearing different values for a feature can also be considered a form of phonological disharmony, the requirement for adjacent phonological elements to have dissimilar features (Leben 1973; Goldsmith 1976; Hansson 2001). Thus, the phonotactics of these ideophones involves Iconic Phonological Disharmony (IPD). Evidence for this pattern of iconicity is mostly limited to the phonotactics of ideophones, with cases involving nonideophonic morpheme alternation being rare. This section presents evidence indicating that such iconicity also exists in tonal alternations targeting ideophones, based on the derivation of verbs in Mwaghavul.

3.1 Deriving verbs from ideophones

Most ideophones in Mwaghavul are adjectives, adverbs or nouns, only a few are un-derived verbs. However, all derived ideophonic verbs in the language have ideophonic nouns, adjectives or adverbs as their bases. The verb formation is morphophonologically marked through one of two lexically determined tonal alternations. This aspect of the Mwaghavul ideophones has not previously been described. Based on the examples in (7–10), we describe verb formation, with a specific focus on the tonal alternations.

- (7) M and M-H tonal overwrites of unreduplicated and reduplicated ideophonic bases

		Verb	Pluractional Verb	
a.	zùt	zūt	zūt-zút	‘how solid substance ejects’
	dĩ:s	dĩ:s	dĩ:s-dĩ:s	‘sound of blood sucking’
	kwàj	kwāj	kwāj-kwáj	‘sound of bubbling hot water’
	vjár	vjár	vjár-vjár	‘pouring of urine’
	fwèr	fwēr	fwēr-fwér	‘how water falls’
b.	ƿǎǎp	ƿǎǎp	ƿǎǎp-ƿǎǎp	‘oily’
	vjàǎp	vjàǎp	vjàǎp-vjàǎp	‘how liquid ejects from a fruit’
	wùláf	wùláf	wùláf-wùláf	‘wide open’
c.	lúfúk	lúfúk	lúfúk-lúfúk	‘smooth and soft’
	wùláf	wùláf	wùláf-wùláf	‘wide open of small thing’

As shown in the second column of (7), the first pattern of alternation involves overwriting every tone of the unreduplicated ideophonic base seen in the first column

with an M tone to derive ‘singular’ verbs. This pattern applies only to a small set of ideophones, consisting of monosyllabic and bisyllabic words. As illustrated in the third column of (7), the second pattern of alternation involves an M-H tonal melody that targets reduplicated ideophones in the derivation of pluractional verbs, such that every TBU of the prefixal/initial reduplicant surfaces with an M tone and every TBU of the base with an H tone.

The M-H pattern of tonal alternation is not limited to the derivation of pluractional verbs but is also involved in the derivation of singular verbs, as seen in the second column of (8).

(8) M-H tonal overwrite of unreduplicated and reduplicated ideophonic bases

	Verb	Pluractional Verb	
a.	bīfɔl	bīfɔl-bīfɔl	‘shabbily dressed’
	kītɪf	kītɪf-kītɪf	‘untidy’
	kɔdɔŋ	kɔdɔŋ-kɔdɔŋ	‘walking effortfully’
	kītʃɔr	kītʃɔr-kītʃɔr	‘old-sounding vehicle’
b.	kɔrjɔŋ	kɔrjɔŋ-kɔrjɔŋ	‘crooked’
	mɔndɔs	mɔndɔs-mɔndɔs	‘big nose’
	vwāplās	vwāplās-vwāplās	‘oversized’
	jālpāt	jālpāt-jālpāt	‘hanging loose’
c.	hāŋlāyāp	hāŋlāyāp-hāŋlāyāp	‘light-weighted’
	hāmhɔyɔf	hāmhɔyɔf-hāmhɔyɔf	‘dryness of cracked feet’
	dʒɛyɛrɛk	dʒɛyɛrɛk-dʒɛyɛrɛk	‘thin/fragile’
d.	gɔkʃirɔk	gɔkʃirɔk-gɔkʃirɔk	‘old or abnormal’
	zɔŋgɪlɔŋ	zɔŋgɪlɔŋ-zɔŋgɪlɔŋ	‘long and thin’
	sālmūtāt	sālmūtāt-sālmūtāt	‘sluggish’
	dīyirík	dīyirík-dīyirík	‘being big and lazy’

To derive singular verbs through the M-H pattern, every nonfinal TBU of the unreduplicated ideophone surfaces with an M tone, while the final TBU surfaces with an H tone, as shown in the second column of (8).² Whether an ideophone undergoes an M or M-H tonal alternation in the derivation of singular verbs is lexically determined, as the alternations are not predictable based on syllable structure or the tone of the unreduplicated ideophones. For example, the unreduplicated ideophones in (7b) and (8a) share the same syllable structure and tone but undergo different tonal alternations: compare (7b) L-L [wùlɔf] ≈ M-M [wūlɔf] versus (8a) L-L [bīfɔl] ≈ M-H [bīfɔl]. Regardless of the tonal alternation seen in the singular form of the second column, all derived pluractional verbs seen in the third column display only the second M-H pattern of tonal alternation. Just like the examples in (7), the ideophones in (8) are reduplicated to derive pluractional verbs, with the reduplicant bearing an M tone on every TBU and the base bearing an H tone on every TBU.

While verbs are derived from both unreduplicated and reduplicated forms of ideophones in (7)–(8), verb formation applies only to the reduplicated form of ideophones in (9). Reduplicating these ideophones – and any ideophones, for that matter – with-

²We refer to verbs derived from unreduplicated ideophones as singular verbs or simply verbs in our discussion, in order to easily distinguish them from pluractional verbs.

out a tonal alternation expresses intensity, iteration or pluractionality, but does not result in a verb formation. To derive pluractional verbs from these ideophones, the tone of the reduplicant and the base must be altered so that the reduplicant bears an M tone on all its TBUs, while every TBU of the correspondent base surfaces with an H tone, as seen in the third column of (9). Thus, only pluractional verbs are derived from this set of ideophones, and the tonal alternation in this case is the same as that of other derived pluractional verbs in the language.

(9) M-H tonal overwrite of derived reduplicated ideophones alone

		Intensity	Pluractional Verb	
a.	dʒùm	dʒùm-dʒùm	dʒùm-dʒúm	‘heavy water plop’
	kùràf	kùràf-kùràf	kùràf-kùráf	‘slow scratching sound’
	kwàs	kwàs-kwàs	kwàs-kwás	‘slow foot-dragging sound’
	tírát	tírát-tírát	tírát-tírát	‘sound of sipping’
b.	dʒúm	dʒúm-dʒúm	dʒúm-dʒúm	‘light water plop’
	kùráf	kùráf-kùráf	kùráf-kùráf	‘fast scratching sound’
	kwás	kwás-kwás	kwás-kwás	‘fast foot-dragging sound’
	wéŋ	wéŋ-wéŋ	wéŋ-wéŋ	‘sound of buzzing’

In (7–9), it is clear that the reduplications are derived. However, reduplication appears to be an inherent property of some ideophones, such as the examples in (10). The language can also derive verbs from these inherently reduplicated ideophones, and the derived verbs in this case are pluractional. Similar to the formation of verbs with derived reduplication, the derivation of verbs from the inherently reduplicated ideophones involves the first string of sounds surfacing with an M tone, and the second string surfacing with an H tone on every TBU. Thus, the first and second strings of the inherently reduplicated ideophones are comparable to a reduplicant and base, respectively. As a result, we can argue that the inherently reduplicated ideophones are historically derived, but the reduplicant and base are synchronically frozen.

(10) M-H tonal overwrite of frozen reduplicated ideophones

		Pluractional verb	
a.	gír-gír	gír-gír	‘sound of a moving truck’
	wùr-wùr	wùr-wùr	‘in a rush’
	tèŋ-tèŋ	tèŋ-tèŋ	‘zigzag’
	zèt-zèt	zèt-zèt	‘hurried walk’
	wùrèt-wùrèt	wùrèt-wùrèt	‘fast walk’
	fùrèt-fùrèt	fùrèt-fùrèt	‘profusely crying’
b.	kípèt-kípèt	kípèt-kípèt	‘flip-flop sound’
	wúléŋ-wúléŋ	wúléŋ-wúléŋ	‘smart look’
	zón-zón	zón-zón	‘describe aimless movement’
c.	ram-ram	rām-rām	‘describes something exciting’

Having exhausted all patterns of verb formation in our dataset, we now turn to the semantics of the derived verbs. In addition to verb formation, the two patterns of verb derivation also introduce various modifications of the lexical meaning of their ideophonic stems. Generally, the derived verbs express that an event (analo-

gously an entity), when compared to other similar events, occurs irregularly, disorderly, unpredictably, haphazardly, sporadically, up and down, nonnormatively, in a zigzag manner, or in other distinguishable ways. For example, the verbs derived from the ideophone [kòrjòŋ] ‘crooked’ in (8b) express that an event or its subcomponent (e.g., walking, dancing, jumping, etc.) is distinguishably crooked when compared to other crooked events. It could also mean that a set of crooked events has a temporal or spatial extension that is distinguishable from another crooked event. Another possible meaning is that the intensity of a crooked event is distinguishable from other crooked events. Similarly, the verb derived from the ideophone [dʒùm] ‘sound of a heavy object dropping in water’ could express that heavy objects fall into a body of water in an irregular manner or produce a distinguishable sound when compared to similar objects dropping into water. The pluractional forms of the derived verbs have the same meaning as their respective singular forms.

Similar to other ideophonic words in the language, the derived verbs have nondisplaceable semantics, so whichever of their possible meanings is adopted depends on the speaker’s intention, and the situational or grammatical context. This is the case regardless of whether they are singular or pluractional. For instance, depending on the context or the speaker’s intention, the verb derived from [dʒùm] ‘sound of a heavy object in water’ could refer to ‘a loud noise emitted as a result of one person (or multiple people) hitting a body of water asynchronously’, ‘one or more big persons jumping into a river at different intervals’, or ‘the sounds made by a person playing in the river being louder than usual’, among other possibilities. The meanings of the derived verbs also suggest positive or negative emotions, depending on the ideophonic stem, the speaker’s intention, or other contextual factors. For instance, the derived verb [wùlɛŋ-wùlɛŋ] ‘to be distinguishably smart-looking’ typically carries a positive affective meaning but can suggest a negative one when used as an insult. On the other hand, the derived verb [dʒùm-dʒùm], meaning ‘to distinguishably sound like a heavy object in water’, typically carries a negative affective meaning but could be positive in some contexts. These features of the derived verbs suggest that they exhibit nondisplaceable and affective meanings, just as expressive morphology and ideophones across languages (Voeltz and Kilian-Hatz 2001; Potts 2007).

The meanings of the derived verbs are comparable to those of ideophones that express distinguishable identity crosslinguistically (Fordyce 1983; Yliniemi 2024; Wivell et al. 2024). This similarity becomes even more apparent when we consider the phonological properties of the derived pluractional verbs. Just like lexicalised ideophones that express distinguishability across languages, the derived pluractional verbs in Mwaghavul involve reduplication, such that each TBU of the reduplicant and its base are assigned different tones. Based on their semantic and phonological similarities to ideophones expressing the distinguishability of identical elements, the derived verbs can also be interpreted as expressing that an instantiation or subcomponent of an event is distinguishable from other similar or related events along a specific dimension. Consequently, the derived verbs convey a distinguishable identity, or more generally, distinguishability.

We found no examples of underived verbs with an M-H tone melody in Mwaghavul, but there are many verbs with M and other tonal combinations in the

language. This observation is based on a list of about six hundred underived verbs and the intuitions of the second author as a speaker of the language. This indicates that the M-H tone melody is reserved for the derivation of verbs. An important question to ask is whether the derived ideophonic verbs pattern like underived verbs in the language, especially in terms of their morphosyntax. In the next subsection, we discuss the morphosyntactic distribution of the derived verbs.

3.2 Morphosyntactic distribution of verbs derived from ideophones

Our goal in this section is to show that the derived ideophonic verbs morphosyntactically pattern like underived verbs in the language. To this end, we compare the morphosyntactic positions of underived verbs and derived verbs. As the reference point of our comparison, we highlight the distribution of underived verbs in (11) before turning to the derived verbs.

(11) Morphosyntactic distribution of underived verbs

- a. *nā:nlè mwà:n ò-kìsūwā*
N. walk LOC-market
'Naanle walked/walks at the market' [PRESENT/PAST]
- b. *nā:nlè m-mwà:n ò-kìsūwā*
N. FUT-walk LOC-market
'Naanle will walk at the market' [FUTURE]
- c. *nā:nlè mū-mwà:n ò-kìsūwā*
N. FOC-walk LOC-market
'Naanle WALKED (did nothing else) at the market' [FOCUS]
- d. *nā:nlè mwà:n fām ò-kìsūwā*
N. walk come LOC-market
'Naanle came here walk(ing)' [SERIAL VERB]
- e. *bà nā:nlè mwà:n ò-kìsūwā kás*
NEG N. walk LOC-market NEG
'Naanle did not walk at the market' [NEGATION]

As discussed in §2.3, the basic word order in Mwaghavul is SVO. We illustrate the morphosyntactic distribution of verbs using the intransitive verb [mwà:n] 'walk' in (11). Present and past tenses are unmarked, so (11a) may be interpreted as either present or past. Future tense is marked with a homorganic syllabic nasal, as shown in (11b). Any word class or phrase can be grammatically focused to signal new or important information. Verbs are focused through a CV reduplicant prefix, where C is a copy of the verb-initial consonant and V is the vowel [i], [u] or [i] with an M tone. As Erin and Fwangwar (2024) suggests, the vowel of the focus marker is determined by the base-initial vowel or consonant. In (11c), we illustrate the focus marking. Verbs can also be combined without a conjunction to form serial verb constructions, as shown in (11d). Negation is doubly marked by the pre- and post-sentential forms [bà] and [kás], as shown in (11e). As shown by the ideophonic verb [kərjòŋ] in (12) that is derived from the ideophone [kərjòŋ] 'crooked', the derived ideophonic verbs exhibit the same morphosyntactic patterns as underived (ideophonic) verbs.

(12) Morphosyntactic distribution of derived verbs

- a. nā:nlè **kōrjój** ò-kìsūwā
N. IDEO.VBZ LOC-market
'Naanle wobbles crookedly at the market' [PRESENT/PAST]
- b. nā:nlè ò-**kōrjój** ò-kìsūwā
N. FUT-IDEO.VBZ loc-market
'Naanle will wobble crookedly at the market' [FUTURE]
- c. nā:nlè kī-**kōrjój** ò-kìsūwā
N. FOCUS-IDEO.VBZ LOC-market
'Naanle **WOBBLED CROOKEDLY** at the market' [FOCUS]
- d. nā:nlè **kōrjój** **vūkmún** ò-kìsūwā
N. IDEO.VBZ IDEO.VBZ LOC-market
'Naanle wobbles crookedly and dusty at the market' [SERIAL VERB]
- e. ní: kī **kōrjój** dʒì
elephant IMP IDEO.VBZ come
'The elephant is coming wobbling crookedly' [SERIAL VERB]
- f. bà nā:nlè **kōrjój** ò-kìsūwā kás
NEG N. IDEO.VBZ LOC-market NEG
'Naanle did not wobble crookedly to the market' [NEGATION]

The intransitive verb [kōrjój] 'wobble crookedly' in (12a) is a derived ideophonic verb. In (12b), the derived verb bears a future-marking prefix. Just like nonideophonic verbs, the derived ideophonic verbs are focused with a CV reduplicant prefix, as shown in (12c). The examples in (12d–e) show that the derived ideophonic verbs can constitute a part or whole of a serial verb. They can also be negated like nonideophonic verbs, as shown in (12f). Regardless of their syntactic context, the derived ideophonic verbs preserve their affective, nondisplaceable and ineffable semantics. In sum, the morphosyntactic distribution of derived ideophonic verbs mirrors that of underived nonideophonic verbs. Although we illustrate their distribution with singular forms, the pluractional counterparts follow the same pattern. Before presenting a formal account of verb formation, we provide descriptive generalisations of the verb formation process in the next section.

3.3 Descriptive generalisations

Based on the data presented above, the following generalisations can be made about the morphophonology of verb derivation from ideophones in Mwaghavul.

(13) Descriptive generalisations

- a. Mwaghavul derives verbs from ideophones through one of two patterns of tonal alternation.
- b. In the first pattern, verbs are derived from unreduplicated ideophonic bases by overwriting every tone of the base TBU with an M tone.
- c. In the second pattern, the verb formation involves overwriting every nonfinal tone of the unreduplicated ideophonic base with an M tone and the final tone with an H tone.

- d. Whether the verb formation from unreduplicated ideophones involves the first or second pattern of alternation is lexically determined.
- e. Regardless of whether the unreduplicated ideophones undergo the first or second pattern of tonal alternation in the verb formation, pluractional verbs are derived from the ideophonic base through a composite of reduplication and the second pattern of tonal alternation.
- f. For the tonal alternation in the derivation of pluractional verbs, every TBU of the reduplicant bears an M tone, and every TBU of the output base bears an H tone.
- g. The two patterns of tonal alternation also introduce the semantic notion of distinguishable identity, modifying the meaning of the ideophonic stems.

The distinction between ideophones and their derived verbs is expressed through tonal alternation, and the pattern of this alternation is lexically specific. An important point to note is that all derived ideophonic verbs in Mwaghavul have ideophonic bases. The ideophonic derived verbs can occur in the same morphosyntactic environments as both underived ideophonic and nonideophonic verbs. In the next section, we account for the source of the tonal alternation and its phonological realisation. Our analysis draws on insights from similar patterns of alternation crosslinguistically.

4 Analysis

We account for the tonal alternations of verb formation in Mwaghavul using Rolle's (2018) framework of grammatical tone, defined as a tonal operation "restricted to a specific morpheme or construction... and not attributable to the general tonal phonology" (Lionnet et al. 2022:386). Hyman et al. (2021) identify two broad categories of grammatical tone. The first type occurs at the word level, where tone is the sole exponent of inflectional or derivational morphology. An example is Kalabari, where verbs are inflected for the imperative mood by concatenating the lexical tones of the verb with an H-L melody that is associated at the right edge of the base, as illustrated in (14). Traditionally, the trigger of grammatical tone in such cases is analysed as a morpheme whose only exponent is a tone or sequence of tones. Under this account, the imperative morpheme of Kalabari has a sequence of H-L tones as its sole exponent.

(14) Imperative marking in Kalabari (Rolle 2018:22)

- | | | | | |
|----|------|----------|------|-----------|
| a. | só | 'go' | sô | 'go!' |
| b. | sò | 'cook' | sǒ | 'cook!' |
| c. | kúró | 'fall' | kúrô | 'fall!' |
| d. | sàkí | 'get up' | sàkî | 'get up!' |

(15) Nominal modification in Kalabari (Harry and Hyman 2014)

- | | | | | | |
|----|------|----------|-----------------|----------|----------------|
| | N | 'this' N | 'the child's' N | 'some' N | |
| a. | námá | mí námá | tùfò námà | jà námà | 'meat' |
| b. | pùlò | mí pùló | tùfò pùlò | jà pùlò | 'oil' |
| c. | bélé | mí bèlé | tùfò bèlè | jà bèlè | 'light' |
| d. | gàrí | mí gàrí | tùfò gárí | jà gárí | 'garri (food)' |

The second type of grammatical tone occurs at the phrase level, as illustrated by the Kalabari examples in (15). When nouns are modified by the proximal demonstrative [mí], they surface with a fixed L-H tonal melody, regardless of their lexical tone. As the possessum of an associative construction, nouns surface with an H-L melody, and with the indefinite quantifier [jà] ‘some’, they surface with an L tone on every TBU. Depending on other evidence in the system, the trigger of a phrasal grammatical tone may be the construction itself, a segmentally null morpheme with a tonal exponent, or the morphosyntactic complement of such a construction. Regardless of what is considered the trigger, the morphosyntactically modified nouns are treated as the targets of grammatical tone. Rolle (2018) classifies concatenative grammatical tones, such as the examples in (14), as nondominant, and the replacive ones in (15) as replacive-dominant. Other types of grammatical tone are also discussed in Rolle (2018).

- (16) Relevant components of grammatical tones (Rolle 2018:28–29)
- a. Grammatical tune: the unique tone or set of tones that covaries with the grammatical construction
 - b. Trigger: the morphemes or constructions which license the tonal operation
 - c. Target: the morpheme or morphemes which is the intended undergoer of a tonological operation
 - d. Host: the morpheme or morphemes on which the grammatical tune appears
 - e. Valuation window: the portion of the target-host which is evaluated with respect to whether its TBUs are valued or unvalued; this can be coextensive with the target-host, or strictly a local subconstituent

Rolle (2018) also proposes a formal framework and a set of terms to capture the phonological and morphosyntactic typology of grammatical tone across languages. Some components of grammatical tone within his framework are highlighted and defined in (16). In this section, we account for the alternations in Mwaghavul using the framework and terminology proposed by Rolle (2018).

4.1 Verbalisation in Mwaghavul involves grammatical tones

We have shown that Mwaghavul derives verbs from ideophones through two patterns of tonal overwrite. The first pattern involves unreduplicated ideophones surfacing with an M tone on all their TBUs, while the second pattern involves unreduplicated ideophones surfacing with an M tone on every nonfinal TBU and an H tone on the final TBU. The derivation of pluractional verbs involves a combination of ideophone reduplication and the second pattern of tonal alternation. In this case, the reduplicant bears an M tone on all its TBUs, while the following base bears an H tone on every TBU. Regardless of their tone melody, all derived ideophonic verbs express the semantic notion of distinguishable identity. The alternations are therefore consistent with the notion of grammatical tone.

We begin our analysis of both alternations by focusing on the derivation of verbs from unreduplicated ideophones. The discussion shows that whether verb formation involves an M or M-H tonal overwrite is lexically specific. Given that the two patterns

of tonal alternation have the same meaning, we assume that the trigger of the tonal alternations is a verbaliser morpheme with two lexically specific suppletive allomorphs, each expressed solely through tone melody. As presented in (17), the first verbaliser has an M tone as its sole exponent, and the second verbaliser has a sequence of M and H tones as its only exponent. The tone sequence of the M-H alternation cannot be explained based on syllable, segment or tone types of the ideophonic base. Since there are no H-M derived verbs, we assume that the linear order of tones belonging to the M-H verbaliser is set in the input.

- (17) Suppletive allomorphs of the verbaliser and their tonal exponents in Mwaghavul
- a. VERBALISER AFFIX 1: M
 - b. VERBALISER AFFIX 2: M H

We now turn to the phonological realisation of the verbalisers, starting with the derivation of singular verbs. The target and host of the M-tone verbaliser is a root morpheme, as M-tone overwriting applies only to unreduplicated ideophones. In this case, the grammatical tone could have been realised on a single TBU of the host, as illustrated in (18a–b). However, the actual realisation and full extension of the grammatical tone of the M-tone verbaliser, as shown in (18c), suggest that its valuation window comprises every TBU of the host – that is, the entire root morpheme. The dashed line indicates the association of a grammatical tone with a host's TBU. Similar to the M-tone verbaliser, the M-H verbaliser also has an ideophonic root morpheme as its target and host. However, the valuation window for each tone of the M-H verbaliser appears to differ from that of the M-tone verbaliser. Specifically, the M tone of the M-H verbaliser seems to span every nonfinal TBU, while the H tone targets the final TBU, as shown in (19). Although this pattern holds for unreduplicated ideophones, our hypothesis is that each tone of the M-H verbaliser also has every TBU of the host as its valuation window. If this is the case, we must address why one tone of the M-H verbaliser does not dominate the other, as illustrated with the unattested forms of (19a–b). One possibility is that there is a requirement for every tone of a grammatical tune to be realised in order to distinguish derived verbs from underived ones (similar to Realize-Morpheme in Gnanadesikan (1997) and Kurisu (2001)). That the entire root morpheme serves as the valuation window for each tone of the verbalisers means that the optimal realisation of the M-H verbaliser in the derivation of singular verbs is not perfect, such as the attested form in (19c). To support the claim that the valuation window of the M-H verbaliser is the entire root morpheme, we turn to the tonal alternation found in the derived pluractional verbs.

- (18) Singular verb: valuation window of the M grammatical tone

- a. * M L
 ⋮ |
 wu laf
- b. * L M
 | ⋮
 wu laf

- c.
- ```

 M
 / \
 wu laf

```
- (19) Singular verb: valuation window of the M-H grammatical tone
- a. \* M
- ```

      M
     / \
  bi    ʃɔl
  
```
- b. * H
- ```

 H
 / \
 bi ʃɔl

```
- c. M H
- ```

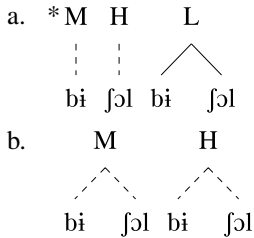
      M H
     / |
  bi  ʃɔl
  
```

As shown in the previous section, the derivation of pluractional verbs from ideophones involves reduplication and an M-H tonal melody. To account for this derivation, two possible analyses can be proposed. The first is that pluractionality is marked by a composite of reduplication and an M-H tonal melody. So, in addition to the verbalisers in (17), this account would require positing a pluractional verbaliser with both reduplication and grammatical tone as its coexponents. The second possibility is that the derivation of pluractional verbs from ideophones involves two morphemes: (i) the M-H verbaliser in (17b) derives the verbs; (ii) reduplication marks pluractionality. Under this account, the semantic similarity between the singular verbs derived through an M-H melody and the derived pluractional verbs can be attributed to sharing the same verbaliser. This account is consistent with the fact that the language has an independent process of marking pluractionality through reduplication (e.g., [pūs rāndōŋ nī] ‘hit the cow’ vs [pūs pūs rāndōŋ nī] ‘hit the cow multiple times’). It is also in line with the crosslinguistic tendency to associate plurality or pluractionality with reduplication (Hurch 2005; Inkelas and Zoll 2005). Based on crosslinguistic and language-internal evidence, we adopt the second possibility to account for the tonal alternation and reduplication in the derivation of pluractional verbs from ideophones. This entails that reduplicated and unreduplicated ideophones can both serve as the target of the M-H verbaliser. We assume that reduplication, as a strategy for marking pluractionality in Mwaghavul, involves morphological doubling (Inkelas and Zoll 2005), such that the derived pluractional verbs contain two root morphemes.

We now turn to the realisation of the M-H verbaliser in the derivation of pluractional verbs. Our account is that one root morpheme and its TBUs are the respective host and valuation window of each tone belonging to the M-H verbaliser, although the two tones of the M-H verbaliser are realised on one root morpheme in the derivation of singular verbs. This means that the optimal realisation of the M-H melody in the derivation of singular verbs is imperfect, as a single host must bear both tones. Similarly, realising the tones of the M-H verbaliser only on the reduplicant or the base in the derivation of pluractional verbs is also not optimal and perfect, as in the

unattested form in (20a). In contrast for the derived pluractional verbs, the number of hosts matches the number of tones, allowing a perfect optimal realisation of the M-H melody, as shown in (20b). In this case, the M tone of the M-H verbaliser is realised on every TBU of the reduplicant and the H tone is realised on every TBU of the base.

(20) Pluractional verb: valuation window of the M-H grammatical tone



To motivate the realisation of the M-H verbaliser in the derivation of pluractional verbs, we compare the phonological properties of the derived pluractional verbs with those of ideophones expressing distinguishable identity across languages. The derived pluractional verbs in Mwaghavul and ideophones expressing distinguishability across languages involve reduplication, such that corresponding segments of the reduplicant and the base are assigned different feature values (e.g., *zigzag* and *pishposh* in English). When the difference involves tone, every TBU of the reduplicant tends to bear a tone that differs from every tone of the base (e.g., [dʒàgà-dʒágá] ‘zigzag/disorderly’ and [màgò-mágó] ‘scam’ in Nigerian Pidgin, Akinbo 2025). In other words, one feature value is associated with the reduplicant and another feature value is associated with the base. The association of opposite feature values with a phonological form and its reduplicant in expressing the distinguishability of identical elements is a pattern of iconicity (Dingemanse and Thompson 2020; Yliniemi 2024). We propose that the realisation of the tones belonging to the M-H verbaliser is also motivated by this pattern of iconicity, such that each tone of the verbaliser gets its own root morpheme and valuation window, rather than share with another tone.

In general, the tonal alternations of the verb formation are triggered by segmentally null verbalisers that have M and M-H grammatical tones as their respective exponents. While the verbaliser with the M tone targets only unreduplicated ideophones, the verbaliser with the M-H grammatical tone targets both unreduplicated and reduplicated ideophones. Each tone of the grammatical tone melody has a root morpheme as its host and every TBU of the host as its own valuation window. The tonal alternations of verb formation in Mwaghavul are consistent with the typology of grammatical tone categorised as replacive-dominant (Rolle 2018). We have also argued that iconicity motivates the realisation of the M-H verbaliser, particularly in the derivation of pluractional verbs. In the next section, we explore whether the derived verbs also share other properties with ideophones across languages.

4.2 Expressiveness and grammatical integration of ideophones

Expressive properties of ideophones include affective meaning, ineffable semantics, foregrounding, marked linguistic features and iconicity (Zwicky and Pullum 1987;

Potts 2007; Fortin 2011). Given that ideophones tend to be morphosyntactically optional and occur outside predicate structure, in the literature they are considered to have a lower degree of grammatical integration (Zwicky and Pullum 1987; Andrason and Bernd 2023). Moreover, when ideophones are grammatically integrated, the degree to which they are integrated into grammatical structure inversely correlates with their degree of expressiveness (Dingemanse and Akita 2017). Therefore, grammatically integrated ideophones are predicted to lose their expressive features or become de-ideophonised. In Mwaghavul, where adverbial, adjectival and nominal ideophones are integrated into grammar through verbal derivation, the prediction is that the derived ideophonic verbs should be less expressive than their ideophonic bases. In this section, we will show that this prediction does not completely hold for the derived verbs in Mwaghavul.

We have categorised the meanings introduced by the verb formation as ‘distinguishable identity’, but the exact meaning varies depending on the ideophonic stem and the discourse context. The meanings also evoke either positive or negative associations. This is likely because ‘distinguishability’ implies nonconformity or irregularity, which may explain the tendency for ideophones expressing distinguishability to appear in insults or praise poetry (Samarin 1969; Dingemanse and Thompson 2020). The affective, variable, and nondisplaceable meanings suggest that the derived verbs are expressive (See §3.1).

The playful and performative nature of ideophones is tied to the expressive property of foregrounding, which is “the use of linguistic forms in such a way that attracts attention” (Havránek 1964:10; Dingemanse and Thompson 2020:205). Strategies of foregrounding across languages include lexical choice, prosody and structural markedness. The derived Mwaghavul verbs also show structural markedness, which are phonotactic patterns and structures that make them stand out from other verbs in the language. For instance, the derived verbs comprise the majority of verbs with three or more syllables and are also the only verbs with an M-H tonal melody. In addition, a reduplicant and its base bearing dissimilar tones can be considered an instance of tone polarity, which is also phonologically marked. These properties indicate that the derived verbs in Mwaghavul are foregrounded through structural markedness.

The iconicity of ideophones across languages is often associated with their structural markedness (Voeltz and Kilian-Hatz 2001; Hinton et al. 1994; Körtvélyessy and Štekauer 2024). This is also the case in Mwaghavul. As shown earlier, realising each tone of the M-H verbaliser on separate root morphemes in the derivation of pluractional verbs involves iconicity, which is a form of nonarbitrary form-meaning mapping and a prominent expressive property of ideophones (Dingemanse et al. 2015). The fact that the M-H tone melody is attested only in derived verbs also points to systematicity, another form of nonarbitrariness that involves “a statistical relationship between the patterns of sound for a group of words and their usage” (Dingemanse et al. 2015:606). The iconicity of the derived pluractional verbs is a crosslinguistic feature of words expressing distinguishable identity. However, the systematicity of the singular verbs derived through the M-H verbaliser is a language-specific feature of Mwaghavul, as we have found no evidence that other Chadic languages exhibit a tonal distinction between derived and underived verbs (Kraft 1970; Newman 2007; Lovstrand 2011).

While all derived ideophonic verbs in Mwaghavul have expressive features, the tone melody of the ones derived through the M-tone verbaliser is neither marked nor iconic. As an argument for the unmarked and arbitrary tone melody of verbs derived with the M-tone verbaliser, we need to compare the singular verbs derived with the M-tone verbaliser to other underived verbs in the language. As illustrated in (21), many underived verbs in Mwaghavul have an M tone. The examples also show that, to derive the gerundive forms, the lexical tones of verbs are overwritten by an M tone. Therefore, the verbs derived through the M-tone verbaliser have the same tonal melody as underived verbs and the grammatical tone of gerundive formation.

(21) Some patterns of gerundive formation in Mwaghavul

	Verb	Gerundive	
a.	wā:m	wā:m	‘to be feeble’
	pīlāj	pīlāj	‘to rinse’
b.	wām	wām	‘to be rotten’
	zūyūm	zūyūm	‘to be generous’

The alternations triggered by verbalisers can overwrite the inherent iconicity of ideophones when that iconicity is encoded in tone. For example, the magnitude iconicity of Mwaghavul ideophones is conveyed through their lexical tones (see §2.3). When such ideophones undergo verb formation, their inherent iconic tone is replaced by tonal systematicity in the derivation of singular verbs through the M-H verbaliser, and by both tonal and segmental iconicity in deriving pluractional verbs. For instance, the magnitude iconicity of the ideophones [wūlāj] ‘wide open’ and [hājłāyáp] ‘light-weighted’ are overwritten by grammatical tones, resulting in the ideophones surfacing as [wūlāj] and [hājłāyáp] in the derivation of singular verbs, and as [wūlāj-wūlāj] and [hājłāyáp-hājłāyáp] in the derivation of pluractional verbs.

In general, the properties of the Mwaghavul derived verbs indicate that they exhibit expressive features, with the exception of systematicity and iconicity, which are attested only in singular and pluractional verbs derived through the M-H verbaliser, respectively. This means that the derived verbs exhibit expressive ideophonic features, and hence can be said to remain within the class of ideophones. The derived verbs also suggest that the grammatical integration of ideophones does not result in a loss of expressiveness in Mwaghavul. In the next section, we present a formal account of the tonal alternation triggered by verbalisers.

4.3 Featural correspondence account of the grammatical tones

We present a formal account of the Mwaghavul grammatical tone alternations in this section. Two main formal accounts have been proposed for phonological alternations with iconic motivations in linguistic theory. One approach is that phonological constraints that operate in expressive morphology are distinct from those of non-expressive morphology (Alderete and Kochetov 2017; Yip 1999). An example of such a constraint is EXPRESS(X) which requires segments in an expressive form to have a specific feature (Alderete and Kochetov 2017). Under this process-based account, the iconicity of an expressive morpheme is solely determined by phonological constraints. Therefore, the prediction is that expressive morphology should be iconic and

different from nonexpressive morphology. This prediction does not hold, given that some patterns of expressive morphology are similar to nonexpressive morphology (e.g., Akinbo 2021a; Molineaux 2026). The alternative approach is that the phonological conditions responsible for nonexpressive alternations can also account for expressive ones (Akinbo 2021a; Molineaux 2026; Akinbo 2025). Under this approach, the expressive alternations involve a phonological item. In this sense, iconicity may arise from either the form–meaning mapping (see Akinbo 2021a; Molineaux 2026) or from phonological constraints acting on the exponent of a morpheme (see Akinbo and Ekiugbo 2024c; Akinbo and Bulkaam 2024a). Our account of the verb formation in Mwaghavul is consistent with the latter approach.

We account for the valuation of the M and M-H verbalisers using morpheme-specific correspondence constraints, as proposed by Finley (2009), within the framework of Optimality Theory (OT) (Prince and Smolensky 2004). The constraints, which were originally proposed for segmental morphemic features, are morpheme-specific instantiations of ANCHOR and CONTIGUITY constraints (McCarthy and Prince 1995). In (22), we present the morpheme-specific versions of the constraints for the M tone of the verbalisers. We assume that H-tone versions of the constraints exist and have the same conditions as the M-tone versions. For brevity, we discuss the operation of the constraints using the M-tone versions.

(22) Morpheme-specific Correspondence Constraints

(Finley 2009; Akinbo 2021a)

- a. LEFT-ANCHOR-M_V: The M tone of a verbaliser must be in correspondence with the left edge of a root.
- b. RIGHT-ANCHOR-M_V: The M tone of a verbaliser must be in correspondence with the right edge of a root.
- c. INTEGRITY-M_V: No M tone of a verbaliser has multiple correspondents in the output.

Our account in §4.1 is that a root morpheme and its TBUs serve as the respective host and valuation window for each tone of the verbalisers. To this end, a root morpheme is formalised as the domain of the constraints. So, if there is a verbaliser in the input, the constraint LEFT-ANCHOR-M_V will require the M tone of the verbaliser to have a correspondent at the left edge of the root morpheme in the output. The constraint RIGHT-ANCHOR-M_V will require the M tone of the verbaliser to have a correspondent at the right edge of the root morpheme. The constraint LEFT-ANCHOR-M_V assigns no violations if the M tone of the verbaliser is realised on the left edge of a root morpheme. Similarly, the constraint RIGHT-ANCHOR-M_V assigns no violations if the M tone of the verbaliser is realised on the right edge of a root morpheme. By associating an M tone to the leftmost and rightmost TBUs simultaneously, each medial TBU automatically bears the same tone, as this is the only phonetically viable option and is consistent with precedence and adjacency relations (Archangeli and Pulleyblank 1994; Pulleyblank 1996; Ní Chiosáin and Padgett 2001). Therefore, the anchor constraints can account for the valuation window of each tone, which spans every TBU of the host. In a situation where the M tone of the verbaliser is realised on a root morpheme but not at the left edge, the constraint LEFT-ANCHOR-M_V assigns a violation for each TBU to the left of the realised M tone. In a similar manner, if the

M tone of the verbaliser is realised but not at the right edge of a root, the constraint RIGHT-ANCHOR-M_V assigns a violation for each TBU to the right of the realised M tone. The constraints LEFT-ANCHOR-M_V and RIGHT-ANCHOR-M_V can be satisfied by realising a copy of the M tone at the left edge and another copy at the right edge. However, this solution will result in a violation of the constraint INTEGRITY-M_V which prohibits the M tone of the verbaliser from having multiple correspondents in the output. If the M tone of the verbaliser is not realised on the TBU of a root morpheme in its input-output mapping, the constraint LEFT-ANCHOR-M_V assigns one violation to each TBU of every root morpheme which is the target of a verbaliser. Similarly, the constraint RIGHT-ANCHOR-M_V assigns one violation to each TBU of every root morpheme which is the target of a verbaliser.

- (23) MAX-Tone (McCarthy and Prince 1995; Zoll 1996): Every input tone has an output correspondent ('Don't delete a tone').

Realising the exponents of the verbalisers can result in the deletion or floating of the ideophones' lexical tones. Without any instrumental evidence, it is difficult to tell whether the lexical tones in this case float or delete. For analytical reasons, we assume that the lexical tones of the ideophones are deleted in the realisation of the grammatical tones. The deletion of the lexical tones can result in a violation of the constraint MAX-Tone in (23), which requires every tone in the input to have a correspondent in the output. So, the constraint MAX-Tone assigns a violation to every tone in the input without a correspondent in the output. By ranking the constraint MAX-Tone below the anchor constraints in (22), we can account for the realisation of the grammatical tones. We assume that the candidates are generated through the concatenation of TBUs with the tone of the verbalisers (similar to Archangeli and Pulleyblank 2022), rather than as a result of infinite generation by GEN as in the standard OT framework (Prince and Smolensky 2004). The account is illustrated in (24–25) for the derivation of singular verbs in the language. The lexical and grammatical tones are distinguished using numeral indexation. The association of the tone with a TBU is indicated with parentheses in place of association lines.

- (24) The M-tone verbaliser and unreduplicated ideophone

	(wùlǎf) ₁ + M _{2V}	INTEG-M _V	L-ANCH-M _V	R-ANCH-M _V	MAX-T
a.	(wùlǎf) ₁ M ₂		*!*	**	
b.	(wùlǎf) ₁		*!*	**	*
c.	(wù) ₁ (lǎf) ₂		*!		
d.	(wū) ₂ (lǎf) ₁			*!	
e. 	(wūlǎf) ₂				*
f.	(wū) ₂ (lǎf) ₂	*!			*

The constraint ranking rules out the candidates in (24a–b) for not realising the exponent of the M-tone verbaliser on a root TBU. For realising the exponent of the verbaliser on only the leftmost or rightmost TBU, the candidates in (24c–d) are ruled out. For realising a copy of the M tone on the leftmost TBU and another copy on another on the right TBU, the candidate in (24f) satisfies the anchor constraints but incurs a fatal violation of the constraint INTEGRITY-M_V. The candidate in (24e) wins

by associating the M tone of the M-tone verbaliser with every TBU of the root morpheme. As shown in Tableau (24), the constraint LEFT-ANCHOR-M_V is ranked above the constraint RIGHT-ANCHOR-M_V, which in turn is ranked above the faithfulness constraint MAX-Tone. To motivate this constraint ranking, we consider the derivation of singular verbs with the M-H verbaliser in (25).

(25) The M-H verbaliser and unduplicated ideophone

		L-ANCH-M _V	R-ANCH-H _V	R-ANCH-M _V	L-ANCH-H _V	MAX-T
	(háŋláyáp) ₁ + M ₂ H _{3V}					
a.	(háŋláyáp) ₁	*!***	*!***	***	***	**
b.	(hāŋlā) ₂ (yáp) ₁		*!***	*	***	*
c.	(háŋláyáp) ₃	*!***		*!***		**
d.	(hāŋlāyáp) ₂		*!***		***	**
e.	 (hāŋlā) ₂ (yáp) ₃			*	**	*
f.	(hāŋ) ₂ (láyáp) ₃			**!	*	*
g.	(hāŋ) ₂ (lá) ₁ (yáp) ₃			**!	**	

As shown in (25), the constraints L-ANCHOR-M_V and R-ANCHOR-H_V are not crucially ranked with respect to each other, but both are ranked above R-ANCHOR-M_V, which in turn is ranked above L-ANCHOR-H_V. The candidates in (25a–d) incur fatal violations of the anchor constraints for failing to realise the M or H tone of the M-H verbaliser. Although both tones of the verbaliser are realised in (25f), this candidate incurs a fatal violation for realising the M tone only on the root-initial TBU. The candidate in (25e) wins for incurring nonfatal violations, as the M and H tones are realised on nonfinal and final TBUs, respectively. The winner and the candidate in (25b) have identical surface tones, but anchor constraints distinguish lexical tones from grammatical tones by privileging association of the high of the verbaliser over association of the high tone of the root.

The realisation of the M-H verbaliser in the winning candidate in (25e), rather than the way it is realised in (25f), is consistent with right-to-left tone mapping in autosegmental theory (Goldsmith 1976; Zoll 2003). Researchers formalise directionality of grammatical tones with markedness constraints such as morpheme-specific alignment constraints (Akinlabi 1996; Zoll 2003). Considering that markedness constraints are not able to distinguish lexical features (e.g., tone) from morphemic or grammatical features, Finley (2009) suggests that the realisation of morphemic features, such as grammatical tones, are better formalised with correspondence constraints. In accord with this suggestion, we have adopted anchor constraints. For example, anchor constraints are able to distinguish the candidates in (25a) and (25b), despite having similar surface tones. In the rest of this subsection, we will show that the same constraint ranking can account for the derivation of pluractional verbs.

The derivation of pluractional verbs involves the M-H verbaliser targeting a morphologically doubled ideophone that expresses plurality. The realisation of each tone of the M-H verbaliser on separate hosts in the derivation of pluractional verbs is

motivated by a crosslinguistic pattern of iconicity in which ideophones expressing distinguishable identity are composed of a reduplicant and a base that bear dissimilar values for a specific feature. The feature value(s) associated with the reduplicant do not cross reduplicant–base boundaries, and the same holds for the feature value(s) associated with the base. This iconic phonological structure can also be captured through how the anchor constraints assign violations. Specifically, if the M tone of the verbaliser is realised on one of the two root morphemes, the anchor constraints in (22) do not assign any violation to the other root morpheme. Similarly, if the H tone of the verbaliser is realised on one of the two root morphemes, the H-tone versions of the anchor constraints in (22) do not assign any violation to the other root morpheme. Based on our account, in which the exponent of the second verbaliser is a sequence of M-H in the input, the realisation of the M tone on the leftmost root and the H tone on the rightmost one preserves the linear order of the tones in their input-output mapping. When there are two root morphemes and the M tone of the verbaliser is not realised on neither root morpheme, the constraint LEFT-ANCHOR-M_V assigns one violation to each TBU of every root morpheme targeted by the verbaliser. In a similar manner, RIGHT-ANCHOR-M_V assigns one violation to each TBU of the two root morphemes. In (26), we show our account of the plural verb formation.

(26) The M-H verbaliser and reduplicated ideophone

	(jàlpàt) ₁ + (jàlpàt) ₂ + M ₃ H _{4V}	L-ANCH-M _V	R-ANCH-H _V	R-ANCH-M _V	L-ANCH-H _V	MAX-T
a.	(jàlpàt) ₁ (jàlpàt) ₂	*!****	*!****	*****	*****	
b.	(jàl) ₁ (pāt) ₃ (jàl) ₂ (pāt) ₄	*!			*	
c.	(jāl) ₃ (pāt) ₁ (jàl) ₄ (pāt) ₂		*!	*		
d. 	(jàlpāt) ₃ (jālpāt) ₄					**
e.	(jāl) ₃ (pāt) ₄ (jàlpāt) ₂			*!	*	*
f.	(jālpāt jāl) ₃ (pāt) ₄				*!	**
g.	(jāl) ₃ (pāt jāl) ₁ (pāt) ₄			*!	*	*

The candidates in (26a–d, e) are ruled out for not realising at least one of the tones belonging to the M-H verbaliser on every TBU of the host. While the candidate in (26f) satisfies the M-tone versions of the anchor constraints by realising the M tone of the verbaliser on every TBU of the first host and the initial TBU of the second host, it incurs a fatal violation of the constraint L-ANCHOR-H_V, because the H₄ is not associated to the left edge of the second conjunct unlike in the winner (26d). The candidates in (26e–g) incur a fatal violation for not realising the M or H tone of the M-H verbaliser on an entire root morpheme. The candidate in (26d) wins by realising the M tone of the M-H verbaliser on every TBU of the first host and the H tone on every TBU of the second host.

As a reviewer suggests, the reduplication and tonal melody of the derived plural-actional verbs could have been analysed in terms of Base–Reduplicant correspondence faithfulness, where “distinguishable identity” could be expressed through tonal dis-

similarity over corresponding syllables in Base–Reduplicant correspondence, such as the analysis of ablaut reduplication across languages (Kenstowicz 1986; Kentner 2017). To determine the tonal specification of the corresponding syllables, this account may also require proposing a fixed M-H melody in the input as a property of the prefixal reduplicant (RED), similar to reduplication with fixed segmentism (McCarthy and Prince 1995; Alderete et al. 1999). However, there is an advantage to analysing the derived pluractional verbs as involving a composite of the M-H verbaliser and a morphological doubling that marks pluractionality. For instance, outside of verb derivation, Mwaghavul pluralises underived verbs through morphological doubling without any tonal alternation (e.g., [pūs] ‘hit’ vs [pūs-pūs] ‘hit repeatedly’). Thus, our account unifies the pluractional marking of derived and underived verbs in the language. Attributing verb derivation and distinguishable identity to the verbaliser also explains the tonal similarity between derived singular and pluractional verbs. Our account of grammatical tone in Mwaghavul is comparable to those of McPherson (2017) and Trommer (2023).

We have shown that the valuation of grammatical tones can be formalised using morpheme-specific anchor constraints. Although the tonal alternations of the derived pluractional verbs are motivated by iconicity, our account shows that we do not need to refer to iconicity in the formalisation of the expressive alternations. As mentioned earlier, these constraints were originally proposed for arbitrary morphemic segmental harmony but have recently been adopted for both segmental and tonal alternations motivated by iconicity (Akinbo 2021a; Akinbo and Bulkaam 2024a; Akinbo 2025). The present work contributes to this line of research by applying these constraints to alternations motivated by iconicity. The successful application of these constraints points to a parallel between iconic and arbitrary phonological alternations. In the next section, we discuss the implications of verb formation in Mwaghavul.

5 Theoretical implications

We have described and analysed the derivation of verbs from ideophones through two patterns of tonal alternation in Mwaghavul. The first involves overwriting every tone of unreduplicated ideophones with an M tone, while the second involves overwriting every tone of unreduplicated ideophones with an M-H melody. Whether an ideophone undergoes the first or second pattern of alternation is lexically determined. To derive pluractional verbs, the ideophones are reduplicated, such that the tones of the reduplicant and its base are overwritten by the M and H tones of the second verbaliser, respectively. The derivation of verbs through tonal alternation is consistent with the notion of grammatical tone (Rolle 2018). In our analysis, lexically specific, segmentally null verbalisers with M and M-H melodies as their respective exponents are considered the triggers of the alternations. While the M-tone verbaliser targets only unreduplicated ideophones, the M-H verbaliser applies to both unreduplicated and reduplicated ideophones. A root morpheme and its TBUs are considered the respective host and the valuation window for each tone of the verbalisers. In addition to verb formation, the alternations introduce meanings that convey the distinguishability of identical elements. Like other ideophones in the language, all the derived verbs

are expressive, but only the singular and pluractional verbs derived through the M-H verbaliser have systematicity and iconicity as their respective properties. Our goal in this section is to explore the implications of these tonal alternations for linguistic theory.

Studies of grammatical tone mostly do not discuss ideophones, even though many of sampled languages are known for integrating ideophones into grammatical structure (Palancar and Léonard 2016; Rolle 2018; Lionnet et al. 2022; Sande 2022). Evidence from verb formation in Mwaghavul suggests that ideophonic words, just like nonideophonic words, can be targets of grammatical tone. Therefore, Mwaghavul contributes to the typology of word classes targeted by grammatical tone. The fact that the derived verbs have the same morphosyntactic distribution as underived nonideophonic verbs indicates that their ideophonic stems are deeply integrated into grammatical structure. The expressive nature of the derived verbs has implications for the view that the degree of grammatical integration inversely correlates with the degree of expressiveness (Akita 2017; Dingemanse and Akita 2017; Andrason and Bernd 2023). This prediction does not hold in Mwaghavul, possibly because ideophonic and nonideophonic words share traditional word classes – such as verbs, nouns, adjectives, and adverbs – in common. Another relevant point is that, unlike the verbalisers in Mwaghavul, the grammatical integration of ideophones in most languages involves nonexpressive morphosyntactic structures. The idea that the properties of a grammatical integrator (i.e., inflectional and derivational morphemes that apply to expressive words) can contribute to the expressiveness of their morphological stems is supported by recent research on alternations motivated by iconicity (Alderete and Kochetov 2017; Akinbo 2021a, 2023a; Akinbo and Ekiugbo 2024c; Akinbo 2025; Molineaux 2026). Therefore, the patterns of verb formation in Mwaghavul support the idea that the relationship between expressiveness and grammatical integration may also be mediated by the grammatical integrator, not merely by the process of integration itself.

Even though ideophones that express the distinguishability of identical elements across languages often exhibit a pattern of disharmony between corresponding segments of the reduplicant and its base, they are rarely mentioned in research on disharmony (Leben 1973; Goldsmith 1976; Kiparsky and Pajusalu 2003). As several studies suggest, this form–meaning mapping is iconic (Fordyce 1983; Dingemanse and Thompson 2020; Yliniemi 2024). Similarly, associating different tones with the reduplicant and its base in the derivation of pluractional verbs in Mwaghavul is consistent with this pattern of iconic disharmony. Traditionally, increased perceptual distinctiveness of individual phonological elements is considered the functional source of disharmony, given that dissimilar adjacent elements are easier to distinguish (Boersma 1998; Frisch et al. 2004). In this case, the traditional notion of perception in phonological theory is understood as purely auditory and limited to phonological form. While the traditional perspective can account for arbitrary disharmony, it is insufficient for iconic disharmonies, given that iconicity involves mapping structural components of real-world elements onto their signifying linguistic forms (Dingemanse 2011a; Emmorey 2014; Punselie et al. 2024). To better capture the integration of iconicity into phonological grammar, we extend the traditional notion of perception beyond auditory and linguistic forms to include other sensory modalities and

real-world elements (following Akinbo 2025; Akinbo and Ekiugbo 2024c). The role of iconicity in this case has an implication for the hypothesis that disharmony is difficult to learn due to its marked articulatory properties (Martin and Peperkamp 2020; Martin and White 2021). Considering that iconicity has been shown to facilitate word learning (Imai and Kita 2014), it is worth investigating whether iconic disharmony is relatively easier to learn than arbitrary disharmony.

Alternations motivated by iconicity are underdocumented, but recent findings from language documentation and description are promising. For instance, iconicity has been shown to motivate syllable metathesis (Akinbo and Ekiugbo 2024c), consonant alternation (Alderete and Kochetov 2017; Molineaux 2026), vowel alternation (Akinbo 2021a) and tonal alternation (Akinbo and Bulkaam 2024a; Akinbo and Ekiugbo 2024c). Notably, a recently described pattern of iconicity in Nigerian Pidgin involves reduplication and tonal alternation in the expression of meanings, which can also be classified as distinguishable identity. Specifically, nonideophonic verbs are reduplicated such that every TBU of the base and its reduplicant are assigned L and H tone, respectively. For example, the words [wàkà-wáká] ‘constantly walking about’ and [hàlà-hálá] ‘constantly yelling about’ are derived from the verbs [wáká] ‘walk’ and [hálà] ‘yell’. The pattern of word formation also applies to a small set of ideophones in the language (Akinbo 2025). This derivational morphology in Nigerian Pidgin is similar to the derivation of pluractional verbs in Mwaghavul, such as [hānjlāyāp-hānjlāyāp] ‘distinguishably light-weighted’ and [vwāplās-vwāplās] ‘distinguishably oversized’ which are derived from [hānjlāyāp] ‘light-weighted’ and [vwāplās] ‘oversized’. Thus, the pattern of verb formation in Mwaghavul contributes to the growing body of work on alternations with iconicity as their functional source. These crosslinguistic patterns of iconic alternations challenge the traditional view that phonological structure is entirely arbitrary (de Saussure 1916; Hockett 1960; Zwicky and Pullum 1987), and instead support a more modern perspective that acknowledges the roles of both arbitrariness and iconicity in grammar (Perniss et al. 2010; Dingemanse et al. 2015).

Crossmodal mapping and the experiential component of iconicity are incompatible with the view that grammar emerges from a language-specific cognitive capacity (e.g., Chomsky and Halle 1968; Chomsky 1986). Moreover, iconic alternations are better explained within a framework that proposes that the capacity to discover and formulate linguistic patterns and constraints lies in domain-general cognitive processes, such as attention to detail, pattern recognition, memory, analogical reasoning and categorisation (Tomasello 2005; Bybee 2010; Archangeli and Pulleyblank 2022). Additional support for the domain-general approach comes from traditions in which linguistic meanings are mapped onto musical melody by imitating phonological features with musical melodies and vice versa (McPherson 2018; Akinbo 2021b, 2023b). Thus, like iconicity, sound-meaning mapping in music also involves crossmodal mapping. Brain imaging studies of iconicity (Lockwood and Tuomainen 2015) and music cognition (Steinbeis and Koelsch 2008; Peretz et al. 2015; Jungers et al. 2016) further show that the brain structures involved in the perception and processing of music, visual imagery, and language significantly overlap. There is also ample evidence suggesting that iconicity is an aspect of natural selection, as plants, animals and other organisms are known to model their behaviour after entities or events in their environments for survival and communicative purposes (Barrett 1987; Launchbaugh and

Provenza 1993; Dalziel et al. 2015; Jamie 2017). This observation also aligns with the perspective that language emerges from domain-general cognitive processes.

(27) The tonal alternation of the possessed noun in an associative construction

	N1	N2		N1 of N2	
a.	lèk	wát	→	lèk wát	‘water of a thief’
	wān	wát	→	wàn wát	‘clay of a thief’
	ḡwó:n	wát	→	ḡwò:n wát	‘pelvis of a thief’
	ḡwó:n	wát	→	ḡwō:n wát	‘surroundings of a thief’
b.	gìr-gìr	wát	→	gìr-gìr wát	‘heavy duty vehicle of a thief’
	rām-rām	wát	→	rām-rām wát	‘excitement of a thief’
	kúráf	wát	→	kúráf wát	‘pot scrapping sound of a thief’

We have explored phonological processes that target only ideophones in Mwaghavul, but many other processes targeting both ideophone and nonideophones are unexplored in the language. For example, Mwaghavul forms associative constructions by conjoining two nouns, where the first noun (N1) is the possessed, and the second noun (N2) is the possessor. In this construction, the first nouns surface with an L tone on all TBUs regardless of their underlying tone, but a subset of high-tone nouns, such as [ḡwó:n] ‘surroundings’ in (27a), surface with an M tone on all TBUs when they occur as the first noun of an associative construction. The ideophonic nouns follow the same general pattern as nonideophonic nouns in the language, except that all high-tone ideophonic nouns surface with an M tone when they occur as the first noun in an associative construction, as illustrated in (27b). Our next step is to focus on the place of ideophones in the analysis of the associative construction and other phonological processes in the language. As phonological alternations targeting ideophones are underdocumented, our future direction is to investigate whether ideophones in other languages are also subject to such alternations. In this context, it is important to determine whether specific patterns of iconic alternation are limited to particular language typologies, geographical areas or language families, or whether alternatively they are found across genetically unrelated languages. The productivity of these alternations and the human intuitions regarding their form–meaning resemblance within and across languages also require experimental testing. Before addressing these and other issues surrounding such alternations, researchers must first document and describe additional patterns of alternation motivated by iconicity across languages.

6 Conclusion

We have investigated whether grammatical tone can target ideophones and whether forms derived from such tonal operations are expressive. By examining the derivation of verbs from ideophones in Mwaghavul, we have shown that grammatical tone can indeed target ideophones. Specifically, the lexical tones of some ideophones are overwritten with an M tone in the derivation of singular verbs. To derive singular verbs from other ideophones, the nonfinal and final lexical tones are overwritten with M and H tones, respectively. Regardless of their pattern alternation in the derivation of

singular verbs, the two classes of ideophones undergo reduplication to mark pluractionality, such that every TBU of the reduplicant and the base bears M and H tones, respectively. In addition to verb formation, the tonal alternations also introduce meanings that convey the distinguishability of identical elements. The triggers of these tonal alternations are analysed as segmentally null allomorphs of a verbaliser, with M and M-H tones as their respective exponents. The derived verbs exhibit expressive features, as well as systematicity and iconicity in the singular and pluractional verbs derived through the M-H verbaliser. The derived verbs occur in the same morphosyntactic environments as the underived verbs. We have also shown that the derived verbs remain within the class of ideophones. This suggests that the expressiveness of grammatically integrated ideophones may be determined not only by the integration process itself but also by properties of the morpheme licensing the integration. The role of iconicity in the derivation of the pluractional verbs challenges the traditional perspective of the complete arbitrariness of phonological grammar, while supporting the more contemporary view that acknowledges the role of both arbitrariness and iconicity in the emergence of grammatical structure. The pattern of verb derivation in the language also contributes to the typology of grammatical tones, especially those with iconicity as their functional source. As Mwaghavul is underdocumented, the present work also contributes to the documentation and description of the language.

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Author Contribution The first and second authors made equal contributions to this work. They were jointly responsible for the conceptualisation of the study, data collection, description and analysis. They also co-drafted and revised the manuscript. All authors read and approved the final manuscript.

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Data Availability The sound files and time-aligned TextGrids of the examples in this work are deposited in an OSF repository, which can be accessed through this link: <https://osf.io/bfpgz/>. The repository also contains the QGIS source files for the language map.

Declarations

Competing Interests The authors state no conflict of interest.

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References

- Akinbo, Samuel Kayode. 2021a. Featural affixation and sound symbolism in Fungwa. *Phonology* 38(4):537–569. <https://doi.org/10.1017/S0952675721000324>.
- Akinbo, Samuel Kayode. 2021b. The language of gáangan, a Yorùbá talking drum. *Frontiers in Communications* 6:1–14. <https://doi.org/10.3389/fcomm.2021.650382>.
- Akinbo, Samuel Kayode. 2023a. Reduplication, repetition and sound symbolism in Fungwa. *Phonological Data & Analysis* 5(2):1–24. <https://doi.org/10.3765/pda.v5art2.72>.
- Akinbo, Samuel Kayode. 2023b. Sound-meaning mapping: Verbal imitation of Super Mario music by Yorùbá gamers. *Language Sciences* 98:1–19. <https://doi.org/10.1016/j.langsci.2023.101553>.
- Akinbo, Samuel Kayode. 2025. Iconicity of grammatical tonal polarity and reduplication in Nigerian Pidgin. *Journal of Linguistics* 61(3):1–26. <https://doi.org/10.1017/S0022226724000276>.
- Akinbo, Samuel Kayode, and Alexander Angsongna. 2025. Relationship between iconicity and grammatical integration as a product of the integrator. In *Dougschrift: A collection of squibs and puzzles presented to Doug Pulleyblank*, eds. Anne-Michelle Tessier, Gunnar Ólafur Hansson, and Kathleen Currie Hall, 1–22. Vancouver: University of British Columbia. <https://arts-pulleyblank-2024.sites.olt.ubc.ca/dougschrift/>.
- Akinbo, Samuel Kayode, and Michael Bulkaam. 2024a. Iconicity as the motivation for the signification and locality of deictic grammatical tones in Tal. *Glossa: A Journal of General Linguistics* 9(1):1–41. <https://doi.org/10.16995/glossa.10666>.
- Akinbo, Samuel Kayode, and Onyinye Anulika Chiemezie. 2024b. Iconicity of grammatical tone polarity in the expression of distinguishable identity. In *Presented at The 21st International Congress of Linguists (ICL) Poznań, Poland (Sept 8-14)*.
- Akinbo, Samuel Kayode, and Philip Oghenesuwohwo Ekiugbo. 2024c. Iconicity as the motivation for morphophonological metathesis and truncation in Nigerian Pidgin. *Open Linguistics* 10(1):1–25. <https://doi.org/10.1515/opli-2024-0013>.
- Akinlabi, Akinbiyi. 1996. Featural affixation. *Journal of Linguistics* 32(2):239–289.
- Akita, Kimi. 2017. The linguistic integration of Japanese ideophones and its typological implications. *Canadian Journal of Linguistics* 62(2):314–334. <https://doi.org/10.1017/cnj.2017.6>.
- Akita, Kimi, and Mark Dingemanse. 2019. Ideophones (mimetics, expressives). In *Oxford research encyclopedia of linguistics*, ed. Mark Aronoff. Oxford: Oxford University Press. <https://doi.org/10.1093/acrefore/9780199384655.013.477>.
- Akita, Kimi, Bonnie McLean, Jiyeon Park, and Arthur Lewis Thompson. 2024. Iconicity mediates semantic networks of sound symbolism. *The Journal of the Acoustical Society of America* 155(4):2687–2697. <https://doi.org/10.1121/10.0025763>.
- Alderete, John, and Alexei Kochetov. 2017. Integrating sound symbolism with core grammar: The case of expressive palatalization. *Language* 93(4):731–766. <https://doi.org/10.1353/lan.2017.0056>.
- Alderete, John, Jill Beckman, Laura Benua, Amalia Gnanadesikan, John McCarthy, and Suzanne Urbanczyk. 1999. Reduplication with fixed segmentism. *Linguistic Inquiry* 30(3):327–364.
- Andrason, Alexander, and Heine Bernd. 2023. On the grammaticalization of ideophones. In *Different slants on grammaticalization*, eds. Sylvie Hancil and Vittorio Tantucci, 237–262. Amsterdam: John Benjamins.
- Archangeli, Diana, and Douglas Pulleyblank. 1994. *Grounded phonology*. Cambridge: MIT Press.
- Archangeli, Diana, and Douglas Pulleyblank. 2022. *Emergent phonology*, Vol. 7. Berlin: Language Science Press.
- Barnabas, Milaham Barnabas. 2011. Origin, migration and settlement: Early history of the Mwaghavul up to 1907. In *Towards a Mwaghavul history: An exploration*, eds. Sale Akila Lohor, Barry Akila Katu, and Umar Habila Dadem Danfulani, 26–51. Bloomington: Xlibris Corporation.
- Barrett, Spencer C.H.. 1987. Mimicry in plants. *Scientific American* 257(3):76–85. <https://doi.org/10.1038/scientificamerican0987-76>.
- Blench, Roger. 2011. Mwaghavul pluractional verbs. In *Papers in Chadic VI: Comparative and descriptive studies*, eds. Doris Löhr and Ari Awaganna, 51–66. Köln: Rüdiger Köppe.
- Blench, Roger. 2013. Mwaghavul expressives. In *Topics in Chadic linguistics vii: Papers from the 6th Biennial International Colloquium on the Chadic Languages, Villejuif, Sep 22-23, 2011*, 53–75. Köln: Rüdiger Köppe.
- Boersma, Paul. 1998. Functional phonology. PhD Thesis, University of Amsterdam.
- Bybee, Joan. 2010. *Language, usage and cognition*. Cambridge: Cambridge University Press.
- Childs, Tucker G. 1994. African ideophones. In *Sound symbolism*, eds. Leanne Hinton, Johanna Nichols, and John J. Ohala, 178–204. Cambridge: Cambridge University Press.

- Childs, George Tucker. 1996. Where have all the ideophones gone? The death of a word category in Zulu. In *Toronto working papers in linguistics*.
- Chomsky, Noam. 1986. *Knowledge of language: Its nature, origin, and use*. New York: Prager.
- Chomsky, Noam, and Morris Halle. 1968. *The sound pattern of English*. New York: Harper & Row.
- Crozier, David H., and Roger M. Blench. 1992. *An index of Nigerian languages*. Dallas: Summer Institute of Linguists.
- Dalziel, Anastasia H., Justin A. Welbergen, Branislav Igic, and Robert D. Magrath. 2015. Avian vocal mimicry: A unified conceptual framework. *Biological Reviews* 90(2):643–668. <https://doi.org/10.1111/brv.12129>.
- de Saussure, Ferdinand. 1916. *Cours de linguistique générale*. Paris: Payot & Cie.
- Dieks, Dennis, and Andrea Lubberdink. 2022. Identical quantum particles as distinguishable objects. *Journal for General Philosophy of Science* 53(3):259–274. <https://doi.org/10.1007/s10838-020-09510-w>.
- Dingemanse, Mark. 2011a. Ezra pound among the Mawu: Ideophones and iconicity in Siwu. In *Semblance and signification*, eds. Pascal Michelucci, Christina Ljungberg, and Olga Fischer, 39–54. Amsterdam: Benjamins.
- Dingemanse, Mark. 2011b. The meaning and use of ideophones in Siwu. PhD Thesis, Radboud University, Nijmegen.
- Dingemanse, Mark. 2019. Ideophone as a comparative concept. In *Ideophones, mimetics, and expressives*, eds. Kimi Akita and Prashant Pardeshi. Vol. 16, 13–33. Amsterdam: Benjamins.
- Dingemanse, Mark, and Kimi Akita. 2017. An inverse relation between expressiveness and grammatical integration: On the morphosyntactic typology of ideophones, with special reference to Japanese. *Journal of Linguistics* 53(3):501–532.
- Dingemanse, Mark, and Bill Thompson. 2020. Playful iconicity: Structural markedness underlies the relation between funniness and iconicity. *Language and Cognition* 12(1):203–224. <https://doi.org/10.1017/langcog.2019.49>.
- Dingemanse, Mark, Damián E. Blasi, Gary Lupyan, Morten H. Christiansen, and Padraic Monaghan. 2015. Arbitrariness, iconicity, and systematicity in language. *Trends in Cognitive Sciences* 19(10):603–615. <https://doi.org/10.1016/j.tics.2015.07.013>.
- Ejeba, Ochala. 2016. *A grammar of Igala*. Port Harcourt: M&J Grand Orbit Communications.
- Emmorey, Karen. 2014. Iconicity as structure mapping. *Philosophical Transactions - Royal Society. Biological Sciences* 369:1–9. <https://doi.org/10.1098/rstb.2013.0301>.
- Erin, Adewumi, and Tongpan Fwangwar. 2024. A grammatical study of reduplication in Mwaghavul. *Journal of The Linguistic Association of Nigeria* 27(1):77–96. <https://doi.org/10.60787/jolan.vol27no1-2.392>.
- Finley, Sara. 2009. Morphemic harmony as featural correspondence. *Lingua* 119(3):478–501. <https://doi.org/10.1016/j.lingua.2008.09.009>.
- Fordeyce, James F. 1983. The ideophone as a phonosemantic class: The case of Yoruba. In *Current approaches to African linguistics*, ed. Ivan R. Dihoff. Vol. 1, 263–278. Dordrecht: Foris.
- Fortin, Antonio. 2011. The morphology and semantics of expressive affixes. PhD Thesis, Oxford University, Oxford, UK.
- Frisch, Stefan A., Janet B. Pierrehumbert, and Michael B. Broe. 2004. Similarity avoidance and the OCP. *Natural Language and Linguistic Theory* 22(1):179–228. <https://doi.org/10.1023/B:NALA.0000005557.78535.3c>.
- Fwangwar, Tongpan Rabo. 2018. Aspects of the phonology of Mwaghavul. MA Thesis, University of Ilorin.
- Gnanadesikan, Amalia Elisabeth. 1997. Phonology with ternary scales. Dissertation.
- Goldsmith, John A. 1976. Autosegmental phonology. Phd, Massachusetts Institute of Technology.
- Graser, Anita, Tim Sutton, and Marco Bernasocchi. 2025. The QGIS project: Spatial without compromise. *Patterns* 6(7):1–8. <https://doi.org/10.1016/j.patter.2025.101265>.
- Hansson, Gunnar Ó. 2001. Theoretical and typological issues in consonant harmony. PhD Thesis, University of California Berkeley.
- Harry, Otelemate G., and Larry M. Hyman. 2014. Phrasal construction tonology: The case of Kalabari. *Studies in Language* 38(4):649–689. <https://doi.org/10.1075/sl.38.4.01hym>.
- Havránek, Bohuslav. 1964. The functional differentiation of the standard language. In *A Prague School reader on esthetics, literary structure, and style*, ed. Paul L. Garvin, 3–16. Washington: Georgetown University Press.
- Hayes, Bruce. 1989. Compensatory lengthening in moraic phonology. *Linguistic Inquiry* 20(2):253–306.
- Heine, Bernd. 2023. *The grammar of interactives*. Oxford: Oxford University Press.

- Hellwig, Birgit. 2011. *A grammar of Goemai*. Berlin: Mouton de Gruyter.
- Hinton, Leanne, Johanna Nichols, and John J. Ohala. 1994. Introduction: Sound-symbolic processes. In *Sound symbolism*, eds. Leanne Hinton, Johanna Nichols, and John J. Ohala, 1–14. Cambridge: Cambridge University Press.
- Hockett, Charles F. 1960. The origin of speech. *Scientific American* 203(3):88–97. <https://doi.org/10.1038/scientificamerican0960-88>.
- Hurch, Bernhard, ed. 2005. *Studies on reduplication*. Berlin: de Gruyter.
- Hyman, Larry M. 1985. *A theory of phonological weight*. Dordrecht: Foris.
- Hyman, Larry M., Hannah Sande, Florian Lionnet, Nicholas Rolle, and Emily Clem. 2021. Sub-Saharan Africa. In *The Oxford handbook of language prosody*, eds. Carlos Gussenhoven and Aoju Chen, 183–194. Oxford: Oxford University Press.
- Ibarretxe-Antuñano, Iraide. 2017. Basque ideophones from a typological perspective. *Canadian Journal of Linguistics* 62(2):196–220. <https://doi.org/10.1017/cnj.2017.8>.
- Imai, Mutsumi, and Sotaro Kita. 2014. The sound symbolism bootstrapping hypothesis for language acquisition and language evolution. *Philosophical Transactions - Royal Society. Biological Sciences* 369:1–13. <https://doi.org/10.1098/rstb.2013.0298>.
- Inkelas, Sharon, and Cheryl Zoll. 2005. *Reduplication: Doubling in morphology*, Vol. 106. Cambridge: Cambridge University Press.
- Jamie, Gabriel A. 2017. Signals, cues and the nature of mimicry. *Proceedings of the Royal Society B: Biological Sciences* 284:1–9. <https://doi.org/10.1098/rspb.2016.2080>.
- Jungers, Melissa K., Julie M. Hupp, and Sara D. Dickerson. 2016. Language priming by music and speech: Evidence of a shared processing mechanism. *Music Perception* 34(1):33–39. <https://doi.org/10.1525/mp.2016.34.1.33>.
- Kenstowicz, Michael. 1986. Multiple linking in Javanese. In *Proceedings of NELS 16*, eds. Stephen Berman, Jae-Woong Choe, and Joyce McDonough, 230–248. Amherst: Graduate Linguistic Student Association.
- Kentner, Gerrit. 2017. On the emergence of reduplication in German morphophonology. *Zeitschrift für Sprachwissenschaft* 36(2):233–277.
- Kiparsky, Paul, and Karl Pajusalu. 2003. Towards a typology of disharmony. *The Linguistic Review* 20:217–241. <https://doi.org/10.1515/tlir.2003.009>.
- Knoefler, Klemens, Jixing Li, Emanuela Maggioni, and Charles Spence. 2017. What drives sound symbolism? Different acoustic cues underlie sound-size and sound-shape mappings. *Scientific Reports* 7(1):1–11. <https://doi.org/10.1038/s41598-017-05965-y>.
- Körtvélyessy, Lívía, and Pavol Štekauer, eds. 2024. *Onomatopoeia in the world's languages*. Berlin: de Gruyter.
- Kraft, Charles H. 1970. Tone in Hausa. *African Studies* 29(2):129–138. <https://doi.org/10.1080/00020187008707324>.
- Kurusu, Kazutaka. 2001. The phonology of morpheme realization. Phd. University of California, Santa Cruz.
- Launchbaugh, Karen L., and Frederick D. Provenza. 1993. Can plants practice mimicry to avoid grazing by mammalian herbivores? *Oikos* 66(3):501–504. <https://doi.org/10.2307/3544945>.
- Leben, William Ronald. 1973. Suprasegmental phonology. PhD Thesis, Massachusetts Institute of Technology.
- Lee, Jin-Seong. 1992. Phonology and sound symbolism of Korean ideophones. PhD Thesis, Indiana University, Bloomington, Indiana.
- Lionnet, Florian, Laura McPherson, and Nicholas Rolle. 2022. Theoretical approaches to grammatical tone. *Phonology* 39(3):385–398. <https://doi.org/10.1017/S0952675723000179>.
- Lockwood, Gwilym, and Mark Dingemans. 2015. Iconicity in the lab: A review of behavioral, developmental, and neuroimaging research into sound-symbolism. *Frontiers in Psychology* 6:1–14. <https://doi.org/10.3389/fpsyg.2015.01246>.
- Lockwood, Gwilym, and Jyrki Tuomainen. 2015. Ideophones in Japanese modulate the p2 and late positive complex responses. *Frontiers in Psychology* 6:1–10. <https://doi.org/10.3389/fpsyg.2015.00933>.
- Lovestrand, Joseph. 2011. The dialects of Baraïn (East Chad). SIL Electronic Working Papers.
- Martin, Alexander, and Sharon Peperkamp. 2020. Phonetically natural rules benefit from a learning bias: A re-examination of vowel harmony and disharmony. *Phonology* 37(1):65–90. <https://doi.org/10.1017/S0952675720000044>.
- Martin, Alexander, and James White. 2021. Vowel harmony and disharmony are not equivalent in learning. *Linguistic Inquiry* 52(1):227–239. https://doi.org/10.1162/ling_a_00375.

- McCarthy, John J., and Alan Prince. 1995. Faithfulness and reduplicative identity. In *Papers in optimality theory*, eds. Jill N. Beckman, Laura Walsh Dickey, and Suzanne Urbanczyk, 249–384. GLSA.
- McPherson, Laura. 2017. Multiple feature affixation in Seenku plural formation. *Morphology* 27(2):217–252. <https://doi.org/10.1007/s11525-017-9300-4>.
- McPherson, Laura. 2018. The talking balafon of the Sambla: Grammatical principles and documentary implications. *Anthropological Linguistics* 60(3):255–294. <https://doi.org/10.1353/anl.2019.0006>.
- Molineaux, Benjamin. 2026. Affective coronal alternations in Mapudungun: Sound symbolism, change and morpho-phonological structure. *Journal of Linguistics* 62(1):35–77. <https://doi.org/10.1017/S002222672500012X>.
- Newman, Paul. 1968. Ideophones from a syntactic point of view. *Journal of West African Languages* 5:107–117.
- Newman, Paul. 2007. Hausa. In *Morphologies of Asia and Africa*, ed. Alan Kaye, 667–711. Winona Lake: Eisenbrauns.
- Ní Chiosáin, Máire, and Jaye Padgett. 2001. *Markedness, segment realization, and locality in spreading*, 118–156. Cambridge: Cambridge University Press.
- Nyang, Stephen, Jacob Bess, and Raymond Dawum. 2013. *Mwaghavul orthography*. Panyam: Mwaghavul Bible Translation Committee.
- Ohala, John J. 1994. The frequency code underlies the sound-symbolic use of voice pitch. In *Sound symbolism*, eds. Leanne Hinton, Johanna Nichols, and John J. Ohala, 325–347. Cambridge: Cambridge University Press.
- Palancar, Enrique L., and Jean Léo Léonard, eds. 2016. *Tone and inflection: New facts and new perspectives*, Vol. 296. Berlin: de Gruyter.
- Peretz, Isabelle, Dominique Vuvan, Marie-Élaine Lagrois, and Jorge L. Armony. 2015. Neural overlap in processing music and speech. *Philosophical Transactions - Royal Society. Biological Sciences* 370:1–8. <https://doi.org/10.1098/rstb.2014.0090>.
- Perniss, Pamela, Robin Thompson, and Gabriella Vigliocco. 2010. Iconicity as a general property of language: Evidence from spoken and signed languages. *Frontiers in Psychology* 1:1–15. <https://doi.org/10.3389/fpsyg.2010.00227>.
- Potts, Christopher. 2007. The expressive dimension. *Theoretical Linguistics* 33(2):165–197. <https://doi.org/10.1515/TL.2007.011>.
- Prince, Alan, and Paul Smolensky. 2004. *Optimality theory: Constraint interaction in generative grammar*. Malden: Blackwell.
- Pulleyblank, Douglas. 1996. Neutral vowels in optimality theory: A comparison of Yoruba and Wolof. *Canadian Journal of Linguistics* 41(4):295–347. <https://doi.org/10.1017/S0008413100016601>.
- Punselle, Stella, Bonnie McLean, and Mark Dingemanse. 2024. The anatomy of iconicity: Cumulative structural analogies underlie objective and subjective measures of iconicity. *Open Minds* 8:1191–1212. https://doi.org/10.1162/opmi_a_00162.
- Rolle, Nicholas Revett. 2018. Grammatical tone: Typology and theory. PhD Thesis, University of California, Berkeley.
- Samarin, William J. 1969. The art of Gbeya insults. *International Journal of American Linguistics* 35(4):323–329. <https://doi.org/10.1086/465077>.
- Sambo, Almustapha Wali. 2016. Canonical shape analysis of Hausa and Bole ideophones. *Macrolinguistics* 5:65–79. <https://doi.org/10.26478/ja2016.4.5.5>.
- Sande, Hannah. 2022. Is grammatical tone item-based or process-based? *Phonology* 39(3):399–442. <https://doi.org/10.1017/S0952675723000106>.
- Steinbeis, Nikolaus, and Stefan Koelsch. 2008. Shared neural resources between music and language indicate semantic processing of musical tension-resolution patterns. *Cerebral Cortex* 18(5):1169–1178. <https://doi.org/10.1093/cercor/bhm149>.
- Thompson, Arthur Lewis, Thomas Van Hoey, and Youngah Do. 2021. Articulatory features of phonemes pattern to iconic meanings: Evidence from cross-linguistic ideophones. *Cognitive Linguistics* 32(4):563–608. <https://doi.org/10.1515/cog-2020-0055>.
- Tomasello, Michael. 2005. *Constructing a language: A usage-based theory of language acquisition*. Cambridge: Harvard University press.
- Trommer, Jochen. 2023. The concatenative structure of tonal overwriting. *Linguistic Inquiry* 55(1):95–151. https://doi.org/10.1162/ling_a_00465.
- Ukaegbu, Nkechi M., Gideon S. Omachonu, and Benita C. Odii. 2017. An autosegmental analysis of Berom tones and tonal processes. *Journal of the Linguistic Association of Nigeria* 20(1):233–242.
- Voeltz, F. K. Erhard, and Christa Kilian-Hatz, eds. 2001. *Ideophones*, Vol. 44. Amsterdam: Benjamins.

- Wallace, Jansson Ted. 2019. A study in flim-flam: Reduplicative ablaut ideophones in English. BA Thesis, Lund University.
- Wescott, Roger W. 1973. Tonal icons in Bini. *Studies in African Linguistics* 4(2):197.
- Williamson, Kay. 1984. *Practical orthography in Nigeria*. Ibadan: Heinemann Educational Books.
- Winter, Bodo, Grace Eunhae Oh, Iris Hübscher, Kaori Idemaru, Lucien Brown, Pilar Prieto, and Sven Grawunder. 2021. Rethinking the frequency code: A meta-analytic review of the role of acoustic body size in communicative phenomena. *Philosophical Transactions of the Royal Society B* 376:1–13. <https://doi.org/10.1098/rstb.2020.0400>.
- Wivell, Grace, Veronica Miatto, Ayla Karakaş, Kalina Kostyszyn, and Lori Repetti. 2024. All about ablaut: A typology of ablaut reduplicative structures. *Linguistic Typology* 28(3):505–536. <https://doi.org/10.1515/lingty-2023-0018>.
- Yakpo, Kofi. 2019. *A grammar of Pichi*. Berlin: Language Science Press.
- Yip, Moira. 1999. Reduplication as alliteration and rhyme. *Glott International* 4(8):1–7.
- Yliniemi, Juha. 2024. Iconicity of modified reduplication in five Himalayan languages. *Linguistics of the Tibeto-Burman Area* 47(2):161–253. <https://doi.org/10.1075/ltba.00024.yli>.
- Zoll, Cheryl Cydney. 1996. Parsing below the segment in a constraint based framework. PhD Thesis, UC Berkeley.
- Zoll, Cheryl Cydney. 2003. Optimal tone mapping. *Linguistic Inquiry* 34(2):225–268.
- Zwicky, Arnold M., and Geoffrey K. Pullum. 1987. Plain morphology and expressive morphology. In *Berkeley linguistics society: Proceedings of the thirteenth annual meeting*, eds. Jon Aske, Natasha Beery, Laura Michaelis, and Hana Filip. Vol. 13, 330–340. Berkeley: Berkeley Linguistics Society.

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